

Research Paper

A top-mounted, chain-driven, bidirectional point absorber type wave energy convertor

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ABSTRACT

The aims of this paper are to introduce the structure of a point absorber type wave energy convertor (WEC) mounted above sea level, to analyze the motion characteristics and power absorptions of such a WEC and to describe the numerous advantages of installing it in the nearshore. The motion equation is solved by representing the excitation force with the dynamic wave pressure and obtaining an approximate system transfer function. The possibility of the buoy becoming airborne or submerged is considered and the power absorption is increased optimally by reshaping the frequency response using a flywheel. The time, phase and spectral averages of power absorptions are presented for offshore waves, nearshore waves and for concentrated nearshore waves using standard wave spectrums and compared. Approximate gains possible with linear and rectangular arrays are also presented. Considering different phase distributions, it is shown that downgrading WECs based on short term power absorptions is not justifiable. With a buoy of radius 2 m, the power absorption of this kind of a WEC is estimated to be about 114 kW in the deep sea and 117 kW in the nearshore, when the energy flux is about 31 kW/m. By concentrating just 13.6 kW/m nearshore waves, it could be increased to 186 kW. If a 3-element longshore in-line array is placed inside the wave concentrator, the total power absorption could be about 556 kW. A 9-element rectangular array would be capable of absorbing about 1 MW by making use of the remaining energy of the down wave.

1. Introduction

The potential of sea waves as an energy source is enormous and the available power in most coastlines is estimated to vary from 5 to 70 kW/m of crest width of the wave (Thurman and Trujillo, 2001; Barstow et al., 2009; Mørk et al., 2010; Pecher and Peter Kofoed, 2017). A good advantage of sea waves is that they are normally predictable a few days ahead (Coe et al., 2021). In addition, seasonal variations are well known except for some unusual storm events. The monthly average of the available wave power in the Southern and Western coasts of Sri Lanka is about 30–40 kW/m in July but that in the same coasts is only about 10–15 kW/m in January (Barstow et al., 2009; Mørk et al., 2010). Normally, such predictions are made using theoretical models of wave characteristics both nearshore and offshore and those models are validated beforehand by comparing with time series of wave data stored in various databases (Barstow et al., 2009; Mørk et al., 2010; Kaselimi and Delikaraoglou, 2018). In these databases, records of decades of sea wave statistics obtained by using satellite altimetry, by using measuring instruments placed in the sea and by visual observations are available. The energy density of waves is much higher than that of winds although the

waves are created by winds and winds are created by the solar radiations. According to (Ringwood, 2008), the wave energy density could be 15 times higher than that of wind for a given amount of solar radiation. Converting wave energy for electricity generation is helpful for developing countries such as Sri Lanka which is dependent on expensive fuel imports for this purpose.

There are mainly four categories of wave energy convertors (WECs) known as wave overtopping devices (Mehlum, 1986; Peter Kofoed et al., 2006; Parmeggiani et al., 2013), oscillating water columns (OWC) (Allsop et al., 2014; Halder et al., 2017), attenuators (Camilla Thomson et al., 2018) and oscillating body systems (OBS). Oscillating body systems include several types of point absorbers (Eriksson et al., 2005; Eriksson et al., 2007; Waters, 2008; Engström et al., 2009; Engström, 2011; Engström et al., 2011; Silvia et al., 2013; Pastor and Liu, 2014; Sjökvist et al., 2014; Brask, 2015; Lejerskog et al., 2015; Rusch et al., 2015; Grüter, 2016; Sinha et al., 2016; Youssef et al., 2016) and wave surge convertors (Watabe, 2008). Reviews on the power absorption and the efficiency of these WEC categories can be found in (Pecher and Peter Kofoed, 2017; Aderinto and Li, 2019). In addition, several variants of horizontal-axis and vertical-axis turbines too have been used for wave energy conversion (Jagath Kumara and Dias, 2015; Jagath Kumara

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Nomenclature			
a	cross sectional area of the buoy	P_T	power absorption of an in-line array
B_{PTO}	damping constant of the PTO	$P(\varepsilon_N)$	time average of power for a given ε_N
c_g	group velocity	S	maximum elongation of the spring
d	draft of the buoy	$S(\omega)$	spectral density of a wave spectrum
D	distance from the shore	s	Laplace variable
E_f	wave energy flux	T	wave period at a given ω
E_{f0}	wave energy flux at deep sea	T_e	energy period of irregular waves
E_{f1}	wave energy flux in the nearshore	T_p	peak period of irregular waves
E_{fmid}	wave energy flux after concentration	T_w	time window for time averaging
e	wave energy density	t	time
F_{hd}	hydrodynamic forces	w_1	width of the concentrator inlet
F_{hs}	hydrostatic forces	w_2	width of the uniform channel
F_{mech}	mechanical forces	x	horizontal coordinate
F_{PTO}	damping force of the PTO	z	vertical displacement of the buoy
F_{rd}	wave radiation force	$\dot{z}$	vertical velocity of the buoy
F_{ss}	spring force	$\ddot{z}$	vertical acceleration of the buoy
F_w	wave excitation force	z_{dm}	position of the buoy down-most point
$F_{w\ max}$	maximum wave excitation force	α, γ, σ	parameters of wave spectrums
g	gravitational acceleration	$\Delta\omega$	frequency resolution in the spectrum
H	wave height	δ	tolerance of buoy position
$H(j\omega)$	frequency response of the WEC	ε	phase of the wave at ω
$H(s)$	system transfer function of the WEC	$\Phi(\omega, h)$	finite depth parameter
H_{m0}	mean significant height	Γ	number of WECs in an in-line array
h	still water depth	ϕ	velocity potential of the wave
I	inertia of the flywheel	η	instantaneous wave elevation
K	index of phase distributions	λ	wavelength at ω
K_h	buoyancy parameter	θ	phase function of $H(j\omega)$
K_s	spring stiffness	ρ	density of sea water
k	wave number	τ_{PTO}	damping torque of the PTO
M	mass of the buoy	ω	angular frequency of the wave
N	index of frequencies in the spectrum	ω_p	peak frequency of the wave spectrum
m_1	mass of the flywheel	ω_r	resonance frequency of the WEC
m_a	added mass	ω_b	lowest unbroken wave frequency
m_n	n^{th} moment of the wave spectrum	CDBD	Chain-Driven Bi-Directional
p_D	wave dynamic pressure	CDBD-FWA	CDBD and Flywheel-Assisted
R	radiation resistance	ER	Estimation Reliability
r	radius of the buoy	JONSWAP	Deep sea wave model
r_1	radii of flywheel and gearwheel	MSL	Mean Sea Level
$P(t, \omega)$	instantaneous power absorption at ω	PA	Point Absorber
$P_{ave}(\omega)$	average power absorption at ω	PTO	Power Take-Off
P_{ave}	spectral average of power absorption	SWL	Still Water Level
$P_{rd}(\omega)$	average radiation loss at ω	TMA	Nearshore wave model
		WEC	Wave Energy Converter

et al., 2018; Ibrahim et al., 2021). In a recent publication, a rotating lift based WEC has been described in (Folley et al., 2023).

In point absorber type WECs, a floating body known as a buoy, moves due to the wave action powering a mechanical system and coupling power to an electrical generator. The wave exerts six forces and moments altogether on the buoy and the buoy moves with six degrees of freedom (Grüter, 2016). However, this movement is normally limited to the heaving motion in the vertical direction (up and down). A well-known example is the direct-driven system in which the buoy is connected to a linear generator, which is fixed to the sea floor, using a rope and a set of springs (Eriksson et al., 2005; Waters, 2008; Engström et al., 2011; Silvia et al., 2013; Sjökvist et al., 2014; Lejerskog et al., 2015).

The objective of the research described in this paper has been to find suitable approaches to increase the usability, increase the life span, reduce the implementation costs and to quantify the power absorption of a point absorber type WEC. Hence, a point absorber type WEC in which the mechanical system and the power conversion module are mounted

above sea level (top-mounted), is considered (Jagath Kumara, 2022). In this top-mounting mechanism, a gearwheel is driven by a chain and the chain is connected to a long spring mounted above water. The mean vertical position of the buoy, which is connected to the other end of the chain, would be automatically adjusted when the mean sea level is changed due to tides. Therefore, this type of a WEC would be functional and usable in most sea conditions. This WEC is also meant for the nearshore so that a simple supporting structure can be erected for mounting it above sea level. Hence, the device would not be vulnerable to heavy storms at offshore. There is no necessity for an expensive under sea power cable and an expensive mooring system used in the deep sea. The other advantages of top-mounting of such a WEC in the nearshore are the protection from ship movements, visibility and easy accessibility from the shore for maintenance, and the safety provided for research and experimental developments. Further, most of the modules of the WEC are not submerged in sea water and there would be less corrosion increasing the life span. Thus, the implementation issues would be effectively simplified and the costs of materials, equipment, installation

and maintenance would be relatively small. Further, this WEC could be placed inside a wave concentrator consisting of a converging channel section, uniform channel section and a diverging channel section used for concentrating, stabilizing and dispersing nearshore waves respectively (Jagath Kumara, 2022). In general, any type of a WEC placed inside such a concentrator would be protected from oblique waves, boat movements and some unusually large waves reaching the shore due to offshore storms. Mooring of WECs would be even more simplified as the supporting structure could be fixed to the walls of the concentrator. Another advantage is that the number of elements in an array would be effectively made smaller because the energy flux is made higher in the uniform channel section by capturing wide wave crests.

In (Eriksson et al., 2005; Engström et al., 2011; Pastor and Liu, 2014; Sjökvist et al., 2014; Brask, 2015; Grüter, 2016), the motion equation of the buoy has been derived by considering the buoy mass, wave excitation force, wave radiation, hydrostatic stiffness, spring stiffness and the damping force of the electrical generator. The parameters representing the wave radiation and the wave excitation force have been found using simulations and then the motion equation has been solved numerically. The buoy position, velocity and the acceleration found semi-theoretically in this way have been used to evaluate the instantaneous and average power absorption for sea states represented by JONSWAP spectrum or by data obtained from real sea trials.

In this paper, the motion equation is derived for the top-mounted configuration by considering the relevant parameters and also the buoyancy imbalance caused by the spring. The wave excitation force is represented explicitly using the dynamic pressure of the wave as a function of the frequency, vertical and horizontal position of the buoy and the time. However, the hydrodynamic coefficients representing the wave radiation are extracted from (Eriksson et al., 2005). Then, the motion equation is represented using an approximate second order transfer function in order to examine the stability of the system, evaluate the frequency response and to improve the power absorption by reshaping the frequency response. The possible range of error caused by the approximation is mentioned. The possibility that the buoy become airborne is checked and the system parameters are adjusted accordingly. Similarly, the buoy submersion possibility is considered too. In this paper, the power absorbed by this kind of a top-mounted WEC is first evaluated for deep sea waves at offshore using the JONSWAP wave spectrum (Hasselmann et al., 1973), just for comparisons. Then, it is evaluated for nearshore waves assuming that it is placed in the shoaling region where the potential energy is still high. In this case, the nearshore waves are modeled using TMA wave spectrum (Kitaigorodskii et al., 1975). Next, the spectrum of the propagating waves inside the uniform channel section of a wave concentrator is formed by concentrating individual components of the TMA spectrum according to (Jagath Kumara, 2022). Then, the power absorption of the WEC placed in concentrated waves is evaluated by using this synthesized wave spectrum. The time averages over short time intervals, the spectral averages and the phase averages of power absorption are found and are compared in certain cases. Estimates of the power absorption of a linear array and a two-dimensional array found using simple calculations are also given in this paper. Although, many different buoy sizes, springs with different spring constants and electrical generators with different amounts of damping have been considered for research, the results for only a selected set is illustrated in this paper. Similarly, monochromatic waves and natural frequencies have been considered but only the results for composite waves are illustrated in detail to avoid it from being too long.

A point absorber type WEC system, which is supported by a spring fixed to an offshore oil and gas platform, has been analyzed using theory and simulations in (Pastor and Liu, 2014). This system is one of the few systems which are found to be somewhat similar to the top-mounted system described in this paper. In (Youssef et al., 2016), a top-mounted WEC implemented using a rack and pinion gear mechanism has been described. In this case, the power absorption has been measured using an experimental prototype under certain wave

conditions. In addition, a number of other types of point-absorbers have been analyzed using theory and/or simulations for various sea states or tested using experimental prototypes installed in the real sea (Eriksson et al., 2005; Eriksson et al., 2007; Waters, 2008; Engström et al., 2009; Engström, 2011; Engström et al., 2011; Silvia et al., 2013; Pastor and Liu, 2014; Sjökvist et al., 2014; Brask, 2015; Lejerskog et al., 2015; Rusch et al., 2015; Grüter, 2016; Thomas et al., 2018; Potapenko et al., 2021; Tan et al., 2021; Murai et al., 2021; Mutombo and Numbi, 2022). Results of these publications are compared to those of the top-mounted WEC described in this paper wherever appropriate. The implementation issues of WECs and WEC arrays in general are considered in (Ringwood, 2008; Faizal et al., 2014; Hong et al., 2014; Sandberg et al., 2016; Felix et al., 2019).

In Section 2 of this paper, the structure of the top-mounted point absorber type WEC proposed by the author is described. In Section 3, the theoretical analysis of the mechanical and electrical systems powered by the wave-buoy interactions is presented. The stability and the frequency response analysis are also included in this section. Then, for simulations and calculations, the offshore waves and the nearshore waves are modelled using discretized versions of JONSWAP and TMA wave spectrums respectively as described in Section 4. The semi-theoretical procedure used for reshaping the frequency response of the WEC system and for evaluating the power absorption is described in Section 5. In Section 6, the estimates of power absorption of the WECs with response reshaping and without response reshaping are presented for JONSWAP waves, TMA waves and for concentrated TMA waves. The reliabilities of these estimates are also given and comparisons are made to other systems whenever possible. In addition, the approximate power absorptions possible with linear and rectangular arrays are illustrated. In Section 7, a discussion on the differences of the methodology and the analysis, system improvements and the figures of merit is given. A conclusion is given in Section 8.

2. The structure of a top-mounted point absorber type WEC

The structure of a chain-driven, bi-directional and flywheel-assisted (CDBD-FWA) WEC is illustrated in Fig. 1(a). The buoy (floater) is connected to a chain and the other end of the chain is connected to an extension spring fixed above the sea level. A gearwheel, which is also mounted above sea level, is driven by the chain. Connected to the same shaft is a flywheel for reshaping the frequency response and decreasing the natural frequency (ω_n) of the whole WEC system. The expansion range of the spring (S) is set to the maximum expected change in the sea level including the amplitude of the expected largest wave. In this way, the length of the chain too must be greater than S . When the wave rises and falls, the buoy moves up and down and the gearwheel rotates clockwise and anti-clockwise respectively due to the wave dynamic pressure and the change in buoyancy. The spring stores energy during the downward motion and releases that energy to help pulling the buoy up during the upward motion. In addition, the spring limits the maximum downward excursion of the buoy. The shaft of a bidirectional and rotatory generator is coupled to the gearwheel either directly or through a gear box as appropriate. Optionally, the spring end can be connected to a linear generator. The polarity of the generated instantaneous voltage, $v(t)$ will be dependent on the upward and downward movements of the buoy. The randomness of this voltage waveform is determined by the randomness of the wave surface, $\eta(t)$. In case of a sinusoidal wave with a constant period, T , $v(t)$ would be sinusoidal with the wave frequency, ω which is $2\pi/T$. With irregular waves, $v(t)$ would be a random function in t . A one suitable option in this case would be to use $v(t)$ to charge a battery bank. The power from the battery bank could be supplied either directly to a DC load, to a DC grid or to an inverter supplying an AC power system.

The buoy is initially at rest with a draft of d at the mean sea level (MSL) when there are no waves, as illustrated in Fig. 1(a) using a dashed line. Due to the wave excitation, at a certain instant, buoy position is

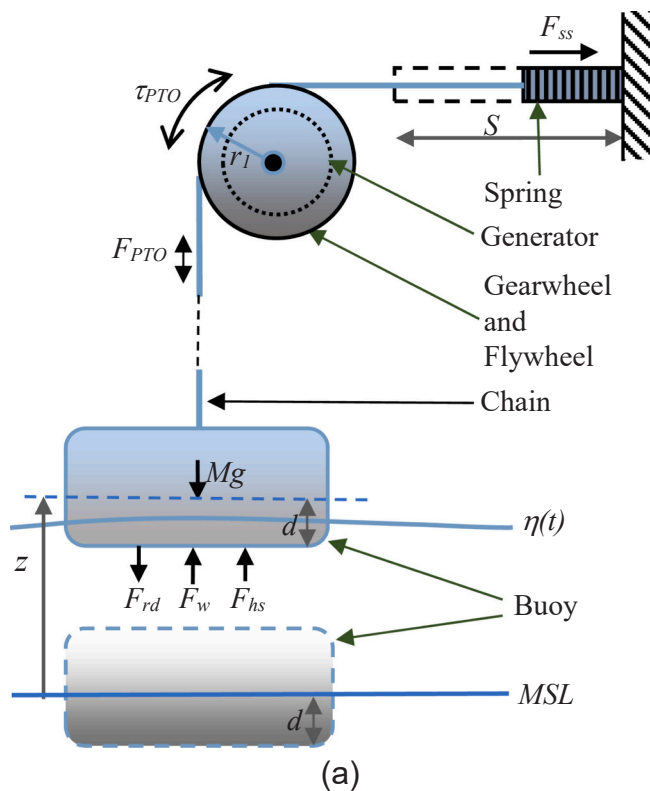


Fig. 1(a). A schematic of a top-mounted point absorber type WEC (not drawn to scale).

shifted by z as illustrated using a solid line. The corresponding forces and torques too are marked in the same figure. At any instant, the power absorbed by the generator is dependent on the damping torque of the generator and the velocity of the buoy movement. In the analysis, a different configuration of this WEC in which there is no flywheel (CDBD WEC) is considered too for comparisons.

In order for the chain to drive the gearwheel properly, the movement of the buoy and the chain must be exactly vertical. However, this may not be the case due to the surge and sway of waves. Therefore, the upper part of the chain should be made to move vertically even though the buoy is free to move not only vertically but also horizontally. Illustrated in Fig. 1(b) is a simple bearing that could be used for this purpose. It is comprised of a set of smaller rings (outer rings) which are free to rotate around a larger ring (inner). In this arrangement, a smooth rope needs to be connected between the chain and the buoy. Then, only the rope is made to move inside the set of outer rings by setting the length of the

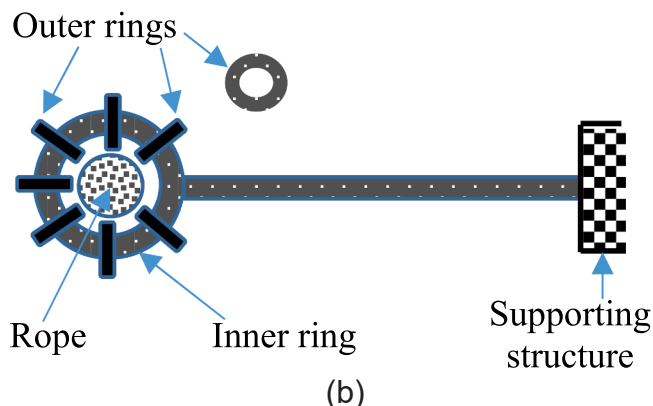


Fig. 1(b). A schematic of a ring bearing (not drawn to scale).

rope section to be greater than S . Therefore, the movement of the chain section, which is in contact with the gearwheel, would always be vertical. Effectively, the forces due to both the surge and the sway would be converted to vertical forces. This is an additional advantage of this kind of a ring bearing (or a motion converter) as more of the wave energy could be captured.

This kind of a WEC is suitable mainly for nearshore installations because it requires a mast or any other structure to support all the modules above sea level. However, there are a number of advantages of such a top-mounted system. Firstly, all the modules of the system including the transmission line is easily accessible for technical staff for installation, testing and maintenance. The whole system would be visible from the shore at all times and problems would be diagnosed very fast. Secondly, the power-take-off (PTO) sub system, comprising the gearwheel, flywheel and the electrical generator with the other electrical components, could be placed inside a cabin which is fixed to the supporting structure. Hence, no additional sealing other than the normal insulations and shields is required for the electrical sub system. The mechanical components such as the gearwheel and the flywheel does not need any special protective coatings. All of these modules are covered from any sea spray or rain increasing their durability. Thirdly, due to those factors, the manufacturing, installation, testing and running costs would be made relatively small. Fourthly, the system would be functional properly despite the fluctuations of the sea level and the wave heights because the spring is extended or shortened accordingly.

Various steps of studies, development and analysis involved in the quantification of the power absorption of the proposed WEC are summarized in the flow chart in Fig. 1(c).

3. Theoretical analysis

In order to find the power absorption of the CDBD-FWA WEC system, the fundamental parameter to obtain is the buoy displacement with time and the buoy velocity. Then, the numerical values of other parameters of interest such as the angular velocity of the gearwheel, hydrodynamic forces and the force of the spring could be found for further analysis. The buoy motion is dependent on the buoy geometry, mass of the moving parts of the system, spring stiffness, flywheel inertia, gearwheel inertia and the damping of the PTO sub system. Although the buoy movement due to the surge and sway too is converted to a vertical motion by using a device such as a ring bearing shown in Fig. 1(b), only the heaving force is considered in the analysis given in this paper. Therefore, the hydrodynamic excitation force of the wave is taken to be due to the dynamic pressure acting on the buoy in the vertical direction (Svendsen, 2006). The hydrodynamic effect of the radiated wave caused by the buoy motion is considered to be a damping force (Eriksson et al., 2005; Engström et al., 2011; Sjökvist et al., 2014; Brask, 2015; Grüter, 2016). The hydrostatic force is due to the buoyancy fluctuations or in other words, the hydrostatic stiffness. The damping torque of the generator is the main source of PTO damping and any other damping effects needs to be minimized to reduce power losses.

The shape of the buoy is taken to be cylindrical with a radius of r in this analysis because the performance of the system with this shape would be favorable as indicated in (Eriksson et al., 2005). At a certain mean sea level (MSL), the buoy would be in equilibrium with a draft of d in the still water if there are no waves. In this case, d is determined by not only the mass and the buoyancy but also the stiffness of the spring. Therefore, considering the buoyancy imbalance,

$$d = \frac{Mg - K_s L}{\rho g a} \quad (1)$$

where M is the mass of the buoy, $a = \pi r^2$ is the cross-sectional area, K_s is the spring stiffness, g is the gravitational acceleration, ρ is the density of sea water and L is $(S - MSL)$. Note that the still water level (SWL) and MSL are taken to be the same in this case because the WEC is considered to be

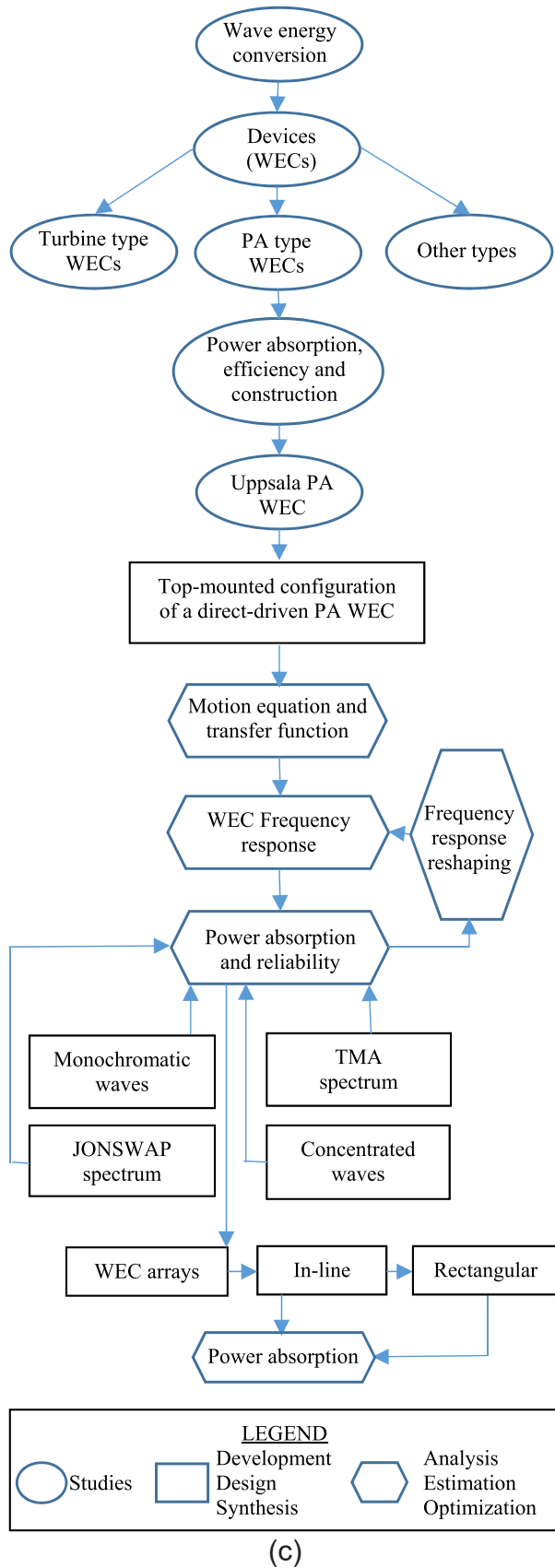


Fig. 1(c). Flow chart of the research described in this paper.

located reasonably seawards from the break point.

When the waves are present, wave-buoy interactions cause the buoy to move with six degrees of freedom. Considering only the vertical motion, which is also known as the heaving motion, if the surface elevation of the waves at time t is $\eta(t)$, let the vertical displacement of the buoy with respect to the equilibrium position be $z(t)$. Then, the vertical velocity and the acceleration are denoted by $\dot{z}$ and $\ddot{z}$ respectively. The main categories of forces acting on the buoy are denoted by F_{hd} (hydrodynamic forces), F_{hs} (hydrostatic forces), F_{mech} (mechanical forces) and F_{PTO} (damping force of the PTO assembly). In this case, the damping torque (τ_{PTO}) of PTO assembly is represented as a force (F_{PTO}) acting on the chain in the vertical direction. The main reason for this is that if the WEC system is connected with a linear electrical generator at the spring end above sea level, the damping force would still be F_{PTO} . However, F_{PTO} could be converted to τ_{PTO} whenever required. Therefore, the foregoing analysis and the results are valid for WEC systems connected either to rotatory generators or linear generators.

The incident waves on the buoy are effectively modified by the wave-buoy interactions. The resulting wave field can be described by the superposition of incident, radiation and diffraction waves and can be modelled using potential theory assuming inviscid, irrotational and incompressible fluid (Eriksson et al., 2005; Brask, 2015; Grüter, 2016). In this case, the diffraction is caused by the deflection and reflection of the incident wave off the buoy. However, if the size of the buoy is small compared to the wavelength, the diffraction could be neglected (McIver and Bennett, 1993; Eriksson, 2005). The drag force on the buoy too is normally very small at low velocities. Accordingly, by using the Laplace equation and the Bernoulli equation, F_{hd} can be represented by the excitation force of the wave (F_w) and the radiation force (F_{rd}) as

$$F_{hd} = F_w + F_{rd} \quad (2)$$

where F_w is due to the wave dynamic pressure given by (Svendsen, 2006),

$$p_D = -\rho \frac{\partial \phi}{\partial t} \quad (3)$$

and ϕ is the velocity potential.

If the buoy heaves in still water, it acts as a wave maker and radiates another wave. According to (Grüter, 2016), the radiation force (F_{rd}) could be derived by considering the radiation potentials for six directions corresponding to the six degrees of freedom and is given by,

$$F_{rd} = m_a \ddot{z} + R \dot{z} \quad (4)$$

where m_a and R are known as the added mass and the radiation resistance. Further details on finding these hydrodynamic damping coefficients can be found in (Malta et al., 2010; Wendt et al., 2017).

The hydrostatic force (F_{hs}) acting on the device is due to the change in buoyancy which is caused by the displacement of the buoy from its previous position. The mechanical forces (F_{mech}) are mainly due to the weight of the buoy (Mg), force of the spring (F_{ss}), inertia of the flywheel (I) and the torques corresponding to the frictional losses of the chain and the bearing losses of the shaft. The frictional and bearing losses are considered to be small in this case because the angular velocity of the shaft is small. In addition, the inertias of the rotor of the generator and the gearwheel should be considered if they are large.

Considering the linear (vertical) motion of the chain, by appropriately converting the relevant parameters of the rotational motion as mentioned before, the damping force of the PTO could be expressed by (Eriksson et al., 2005; Grüter, 2016),

$$F_{PTO} = B_{PTO} \dot{z} \quad (5)$$

where B_{PTO} is the damping constant. However, B_{PTO} is dependent on the current drawn from the generator and in turn on the load impedance connected to the generator (Eriksson et al., 2007; Sjökvist et al., 2014;

Adachi and Takahara, 2016). This is specifically a concern in power systems where the electrical load is unlikely to be a constant and it is directly connected to the generator. Note that according to (Eriksson et al., 2007; Sjökvist et al., 2014), if the imaginary part of the load impedance is small compared to the resistive part and the resistive part is high enough, B_{PTO} could be almost constant. In (Wang et al., 2020), an optimization routine is described to find the suitable load resistance to match several sea states and to maximize the power generation. In this paper, the optimal values of B_{PTO} and the mass of the flywheel (m_f) are found for maximizing the power generation with a predefined level of reliability in a given sea state. The dependence of B_{PTO} on the load is to be taken care of by the effective resistance of the battery bank to which the generator is connected. If the battery bank is an array of a number of units, each unit could be switched on or off to change the effective resistance such that the resulting B_{PTO} is optimal for the prevailing sea state. Similarly, by coupling and decoupling segments of a flywheel, m_f could be changed according to the sea state. Further considerations on the validity of this control method are discussed in Section 7 of this paper.

Then, the equation of motion for the buoy is given by

$$F_w - F_{rd} - Mg + F_{hs} + F_{ss} - F_{PTO} - I\ddot{z}/r_1^2 = M\ddot{z} \quad (6)$$

where, $I = 0.5m_f r_1^2$ and r_1 are the inertia and the radius of the flywheel respectively. Note that the radii of the gear wheel and the flywheel is taken to be the same in this case for simplifying the equations although such similarity is not necessary. The terms due to the mass, hydrostatic stiffness and spring stiffness in Eq. (6) too can be simplified as

$$-Mg + F_{hs} + F_{ss} = K_h \eta - (K_h + K_s)z \quad (7)$$

where $K_h = \rho g a$. Note that in Eq. (7) and in later equations, $\eta(t)$ and $z(t)$ are written as just η and z respectively for simplicity but the time dependence is implied.

By substituting for F_{rd} from Eq. (4) and F_{PTO} from Eq. (5) the motion equation can be written as,

$$(M + m_a + 0.5m_f)\ddot{z} + (R + B_{PTO})\dot{z} + (K_h + K_s)z = F_w + \eta K_h \quad (8)$$

Substituting for ϕ in Eq. (3) from the linear wave theory (Svendsen, 2006),

$$F_w = a p_D = -a\rho \frac{\partial \phi}{\partial t} = \rho g a \eta \frac{\cosh[k(z'+h)]}{\cosh(kh)} \quad (9)$$

where $z' = (z-d)$ is the distance to the point below MSL at which p_D is being evaluated, k is the wave number ($2\pi/\lambda$), λ is the wavelength and h is the still water depth. Note that z' is also the distance to the bottom of the buoy. In Eq. (9), (Svendsen, 2006),

$$\eta = \frac{H}{2} \cos(\omega t - kx) \quad (10)$$

where H is the wave height, $\omega = \frac{2\pi}{T}$ is the angular frequency, T is the wave period and x is the horizontal coordinate. Note that according to the linear wave theory, H is assumed to be very small compared to λ and Eq. (3) for p_D is not valid above MSL. However, p_D may be assumed to be hydrostatic in this case.

In order to find the hydrodynamic coefficients m_a and R , and the wave excitation force F_w in Eq. (8), software simulators such as WAMIT or ANSYS could be used according to (Eriksson, 2005; Silvia, 2013; Brask, 2015; Grüter, 2016). In these cases, m_a and R have been found by simulating the hydrodynamic effects of the buoy which is heaving in otherwise calm water. A useful property of these coefficients is that they are dependent only on the buoy geometry, draft (d) and the angular frequency (ω) of the wave. Therefore, m_a and R illustrated in (Eriksson, 2005; Silvia, 2013) for different r , d and ω could be used to represent the hydrodynamic effects of cylindrical buoys connected to other types of point absorbers too.

After solving Eq. (8) for z using Laplace transforms, numerical methods or any other method, F_{PTO} could be evaluated from Eq. (5). Then, the instantaneous power absorbed by the PTO subsystem at a frequency of ω is given by (Eriksson et al., 2005; Grüter, 2016),

$$P(t, \omega) = F_{PTO} \dot{z} = B_{PTO} (\dot{z})^2 \quad (11)$$

3.1. The system transfer function of the WEC

Rather than trying to solve Eq. (8) for z rigorously, it is more useful to first obtain the transfer function of the WEC system and then to carry out possible analyses and optimizations. Therefore, noting that $\frac{\cosh[k(z'+h)]}{\cosh(kh)}$ in Eq. (9) is unity and also maximum at MSL ($z' = 0$),

$$F_{w \max} = \rho g a \eta \quad (12)$$

Because F_w is decreasing slowly with z' , $F_{w \max}$ could be used in Eq. (8) making the right hand side of it only a function of η . The advantage is that Eq. (8) could be simplified into a second order differential equation in this way and in turn be translated into a second order transfer function. In support of this simplification, it has been found that the variation of F_w is at most 20% over the expected ranges of z and z' . Hence, z represented by this simplification would be an upper bound which is somewhat higher than the real value.

Using Laplace transforms, WEC system transfer function ($H(s)$) could be obtained by assuming that the WEC system was initially at rest (Kamen and Heck, 2000). Therefore, η and $F_{w \max}$ can be represented by a 'sine' function to excite the system starting from zero. In this case, the horizontal placement of the buoy must be $x = (4\beta + 1)\lambda/4$; $\beta = 0, 1, 2, \dots$ such that

$$\eta = \frac{H}{2} \sin(\omega t) \quad (13)$$

By substituting $F_w = F_{w \max}$ in Eq. (8) and taking Laplace transforms of both sides (Kamen and Heck, 2000), the transfer function of this WEC system can be expressed as

$$H(s) = \frac{D_1 + D_2}{As^2 + Bs + C} \quad (14)$$

where

$$D_1 = \rho g a \frac{H}{2} = K_h \frac{H}{2}, \quad D_2 = K_h \frac{H}{2} \quad (15)$$

$$A = M + m_a + 0.5m_f, \quad B = R + B_{PTO}, \quad C = K_h + K_s \quad (16)$$

Note that in the case of CDBD WEC, without a flywheel, m_f in (16) is zero. With this transfer function in Eq. (14) which is of second order too, the stability issues, response to a monochromatic wave and the frequency response could be studied. The vertical position (z), velocity ($\dot{z}$) and the acceleration ($\ddot{z}$) of the buoy could then be found for any ω leading to the forces and torques in the system and the power absorbed by the generator.

3.2. The system stability

The right hand side of the motion equation given in Eq. (8), contains $\eta \times \frac{\cosh[k(z'+h)]}{\cosh(kh)}$ which is a nonlinear term. However, Eq. (8) simplifies to a linear differential equation with the substitution of Eq. (12) as described in the subsequent analysis in Section 3.1. Further, the WEC system parameters do not change with time except for the load connected to the generator. Hence, for a given load and the corresponding B_{PTO} , the system is time-invariant. The wave excitation force and the buoyancy effects are the inputs to the system, which are continuous functions in time. Then, the stability tests described in (Kamen and Heck, 2000) are applicable to this linear, time invariant and continuous-time system. In this way, the transfer function in Eq. (14) represents a stable system if

$A > 0$; $B > 0$ and $C > 0$. Therefore, both CDBD-FWA and CDBD WEC systems are stable because M , m_a , m_1 , R , B_{PTO} , K_h and K_s are always positive. This implies that the buoy heaves safely with bounds under the wave excitation and it would return to the initial equilibrium position when there are no waves.

3.3. The Frequency response of the buoy motion

The frequency response $H(j\omega)$ indicates how the motion response of the WEC system vary when the wave frequency (ω) vary. In this case, by substituting $s = j\omega$ in Eq. (14) (Kamen and Heck, 2000),

$$|H(j\omega)| = \frac{D}{\sqrt{(C - A\omega^2)^2 + (B\omega)^2}} \quad (17)$$

$$\theta = \angle H(j\omega) = -\tan^{-1}\left(\frac{B\omega}{C - A\omega^2}\right) \quad (18)$$

where $|H(j\omega)|$ and θ are the magnitude function and the phase function of the frequency response respectively and $D = D_1 + D_2$. Therefore, the buoy displacement for a given monochromatic frequency of ω is given by (Kamen and Heck, 2000),

$$z = |H(j\omega)|\sin[\omega t + \angle H(j\omega)] \quad (19)$$

Then, $\dot{z}$ and $\ddot{z}$ could be found by differentiating z with respect to t once and twice respectively. Because z and $\dot{z}$ are now sinusoidal, from Eq. (11), the average power absorbed at a given frequency of ω is given by

$$P_{ave}(\omega) = 0.5B_{PTO}|\dot{z}|^2 \quad (20)$$

In this case, this is also the power absorbed by the generator, if the friction and bearing losses are neglected. The electrical power output of the generator would be somewhat less due to the generator losses.

Note that if $D = D_1 + D_2$, the corresponding values of z and $\dot{z}$ are upper bounds due to the use of the $F_{w\max}$ in D_1 . However, the true value would not be much lower than the upper bounds as the deviation of F_w from $F_{w\max}$ is small as explained previously in this Section. If only the buoyancy effect is considered with $D = D_2$, the corresponding values of z and $\dot{z}$ are lower bounds. However, the true values are always higher than these lower bounds as there is always a considerable amount of dynamic pressure whenever there are waves. In this way, either upper bounds or lower bounds of $P_{ave}(\omega)$ could be found using Eq. (20).

4. Sea wave models

In this paper, three different kinds of wave environments representing the offshore (deep sea), nearshore (shallow sea) and the concentrated nearshore waves are considered for evaluating and comparing the power absorption of the top-mounted point absorber type WEC system with and without response reshaping. These WECs are meant for the nearshore but however they are analyzed in the deep sea too for the purpose of comparison. Both in the offshore and the nearshore, waves are irregular in nature and hence they are described using standard wave models. The offshore waves are modelled using JONSWAP model which is characterized mainly by the peak frequency (ω_p), peak period (T_p), mean significant height (H_{m0}), energy period (T_e) and the energy flux (E_p). The nearshore waves are modelled by using TMA model with the same parameters. These two types could be generated using standard formulations (Hughes, 1984; Rueda-Bayona et al., 2020). The concentrated nearshore waves are generated by manipulating TMA waves (Jagath Kumara, 2022).

4.1. JONSWAP wave model

The JONSWAP (Joint North Sea Wave Energy Project) wave model

has been formulated to represent the wave statistics in the North Sea (Hasselmann et al., 1973; Hughes, 1984; Elgar et al., 1985; Rueda-Bayona et al., 2020). It is a modification of the earlier versions of (Phillips, 1958; Pierson and Moskowitz, 1964) which have been formulated to represent fully developed wind generated gravity waves. When the sea waves are fully developed, they are considered to be in a state of equilibrium and any additional input of wind energy would cause wave breaking. The JONSWAP wave spectrum in contrast is a representation of partially developed waves in which case winds are still blowing and the fetch is not long enough for the waves to fully develop (Hasselmann et al., 1973). The spectral density of JONSWAP waves has been specified using four parameters (Hughes, 1984; Rueda-Bayona et al., 2020) by

$$S_{JONSWAP}(\omega) = \alpha \frac{g^2}{\omega^5} e^{\frac{5}{4}\left(\frac{\omega_p}{\omega}\right)^4} \gamma^{\exp\left(-\frac{1}{2}\right)\left(\frac{\omega - \omega_p}{\sigma\omega_p}\right)^2} \quad (21)$$

where the shape of the spectrum at high ω is determined by α and the sharpness of the spectral peak around ω_p is determined by γ . The values of α and γ are dependent on the wind speed and the fetch. In this way,

$$\alpha = \left[1 - 0.287 \ln(\gamma)\right] \left(\frac{5}{16g^2 H_{m0}^2 \omega_p^4}\right) \quad (22)$$

Normally,

$$\gamma = 3.3 \text{ and } \sigma = \begin{cases} 0.7; & \omega \leq \omega_p \\ 0.9; & \omega > \omega_p \end{cases} \quad (23)$$

which are typical for growing wind seas in deep sea.

By definition, the phases (ϵ) of each ω are random variables that are statistically independent and identically distributed. Each individual distribution is uniform between 0 and 2π .

4.2. TMA wave model

The wave hydrodynamics in the nearshore are much more complex, which involve several processes such as refraction, reflection, deflection, diffraction and dissipation (Svendsen, 2006). Basically, when the wave propagates into the nearshore where $h < \lambda/2$, the wavelength decreases while the wave height increases due to the refraction caused by the decreasing water depth. These shoaling waves are made to break when the decrease in λ is too small and/or the increase in H is too high compared to h depending on the wave steepness at deep sea which is denoted by (H_0/λ_0) . After breaking, the shape of the wave is continually transformed into that of a bore while propagating towards the shore and dissipating energy (Svendsen, 1983; Basco and Yamashita, 1987; Bendykowska and Werner, 1988; Hansen, 1990; Svendsen, 1992; Sørensen et al., 1994; Svendsen, 2006; Goda, 2010; Power et al., 2010; Inch, 2014). However, the wave conditions in the deep sea and the shallow sea are similar in some way considering the fact that some waves could break in the deep sea. The deep sea is not so deep with respect to longer waves and the shallow sea is not so shallow with respect to shorter waves. Although, such long waves with a wave length exceeding hundreds of meters and a wave period exceeding 20 s are not frequent normally, the implication is that the mathematical formulations of both the deep sea model and the nearshore model could be similar (Hughes, 1984; Bergdhal, 2009). Hence, JONSWAP spectrum has been further modified by (Kitaigorodskii et al., 1975) to obtain the shallow water spectrum known as the TMA spectrum (Hughes, 1984) where

$$S_{TMA}(\omega) = \Phi(\omega, h) S_{JONSWAP}(\omega) \quad (24)$$

and

$$\Phi(\omega, h) = \frac{k^{-3} \left(\omega, h \right) \frac{\partial k(\omega, h)}{\partial \omega}}{k^{-3} \left(\omega, \infty \right) \frac{\partial k(\omega, \infty)}{\partial \omega}} \quad (25)$$

In Eq. (24), the finite depth effects of nearshore waves are represented by $\Phi(\omega, h)$ term. An iterative formula for evaluating it with some approximations is given in (Hughes, 1984). This model has been tested using time series of wave data collected in three beaches. The letters T, M and A in TMA are respectively the first letters of the names of these three beaches.

4.3. The concentrated TMA spectrum

In the nearshore, more energetic waves with higher energy flux than normal could be realized using a wave concentrator (Sørensen et al., 1994; Dupont et al., 2012; Maliyadda et al., 2014; Jagath Kumara and Dias, 2015; Szmidt and Hedzielski, 2016; Moulton et al., 2017; Jagath Kumara et al., 2018). In this case, the available wave energy in wide wave fronts are converged to a narrow open channel in order to either reduce the lateral dimension of a WEC unit, reduce the number of elements in a WEC array or to make them function even in weak wave conditions. However, long waves with long periods are rejected by open channels depending on the width (Akylas and Mei, 2001–2014; Robert, 2024; Jagath Kumara, 2022). This may be first seen as a disadvantage because the average energy flux over the spectrum of waves are effectively decreased. However, such long waves are relatively infrequent and less powerful. For example, in TMA wave spectrum, a rapid decrease in the spectral density could be observed for $\omega < \omega_p$. In addition, filtering of such wave components in this way could be useful for simplifying the control systems of some WECs. Besides, very long waves which could be hazardous are prevented from entering the concentrator, safeguarding the WEC systems placed inside.

The parameters of a monochromatic wave propagating inside the concentrator such as the wave number, wavelength, phase velocity and the group velocity could be evaluated according to (Akylas and Mei, 2001–2014; Robert, 2024) at a given point. The wave height variation along the concentrator could be evaluated by considering the refraction (Svendsen, 2006) and the level of convergence or divergence of channels (Chang et al., 1979). These results could then be combined to find the energy density and flux for a given wave frequency (Jagath Kumara, 2022). Further, some of the propagating waves inside the concentrator could be broken if they are still too long, drastically reducing the wave height and the energy. In this case, some of the relevant parameters such as the point of breaking, the wave-height-to depth ratio (H/h) and then the wave energy flux could be estimated as described in (Jagath Kumara, 2022) by combining the empirical results and empirical formulations given in (Svendsen, 1983; Basco and Yamashita, 1987; Hansen, 1990; Svendsen, 1992; Svendsen, 2006).

The spectrum of the incoming waves to the concentrator can be represented using the TMA spectrum ($S_{TMA}(\omega)$) but that of the concentrated waves ($S_{concentrated}(\omega)$) is unknown. This spectrum may be formed by evaluating the wave parameters at the required point inside the concentrator according to (Jagath Kumara, 2022) for each incoming TMA wave frequency (ω) individually. In this paper, $S_{concentrated}(\omega)$ is formed for the middle of the uniform channel section (halfway through the channel), where the point-absorber type WECs are expected to be placed.

4.4. Discretization of the wave spectrums

It is possible to calculate $z(\omega)$, $\dot{z}(\omega)$ and $P_{ave}(\omega)$ of CDBD WECs by using the available energy ($S(\omega)d\omega$) at a certain frequency ω where $S(\omega)$ is a given wave spectrum. However, for example, in order to find the total P_{ave} , it is necessary to find $\int P_{ave}(\omega)d\omega$ over the continuous spectrum of frequencies. This is not feasible because the frequency

dependent parameters such as m_a and R are not available in functional form. Therefore, the spectrum has to be discretized with an acceptable level of frequency resolution ($\Delta\omega$) (Elgar et al., 1985; Pastor and Liu, 2014). On this regard, it has been shown in (Pastor and Liu, 2014) that even 20 discrete frequency components would be adequate for representing the JONSWAP spectrum. In this case, a good agreement of numerical calculations and simulation results has been observed.

In the analysis described in this paper, 24 frequency components (ω_N , $N = 1, 2, \dots, 24$) with 24 different phases (ε_N , $N = 1, 2, \dots, 24$) selected randomly are used to represent each spectrum with a resolution ($\Delta\omega$) of 0.1 rad/s. Therefore, if e_N is the energy density of the N^{th} frequency component (ω_N) in general, (Elgar et al., 1985; Pastor and Liu, 2014),

$$S(\omega_N)\Delta\omega = \frac{e_N}{\rho g} \quad (26)$$

Then, the height and the instantaneous elevation of this wave component are given respectively by

$$H_N = 2\sqrt{2S(\omega_N)\Delta\omega} \quad (27)$$

$$\eta_N(t) = \frac{H_N}{2} \cos(\omega_N t - k_N x + \varepsilon_N) \quad (28)$$

where k_N is the wave number of the N^{th} component. Hence, for example, the total elevation of an irregular wave described by a given wave spectrum at a horizontal position of x in the direction of propagation and at a time instant of t is given by (Elgar et al., 1985; Pastor and Liu, 2014),

$$\eta(x, t) = \sum_{N=1}^{24} \eta_N(t) \quad (29)$$

A sea state of a given wave spectrum is considered to be defined by the particular set of ω_p , α and h , of which e_N , H_N and $\eta(x, t)$ could be found in this way. Further, if the group velocity of the N^{th} wave component is $c_g(\omega_N)$, the energy flux of this sea state is given by (Svendsen, 2006),

$$E_f = \sum_{N=1}^{24} c_g(\omega_N) e_N = \sum_{N=1}^{24} c_g(\omega_N) \rho g S(\omega_N) \Delta\omega \quad (30)$$

The mean significant height and the energy period could be evaluated statistically using the moments of the wave spectrum as (Brask, 2015; Grüter, 2016),

$$H_{m0} = 4\sqrt{m_0}; \quad T_e = 2\pi m_{-1}/m_0 \quad (31)$$

where the n^{th} moment is given by

$$m_n = \int_0^\infty \omega^n S(\omega) d\omega \quad (32)$$

For a discretized spectrum, m_0 and m_{-1} could be approximated by (Pastor and Liu, 2014; Brask, 2015; Grüter, 2016)

$$m_0 = \sum_{N=1}^{24} S(\omega_N) \Delta\omega; \quad m_{-1} = \sum_{N=1}^{24} \frac{S(\omega_N)}{\omega} \Delta\omega \quad (33)$$

Different sea states of JONSWAP and TMA spectrums could be obtained by changing ω_p and α . For example, a JONSWAP sea state with $\omega_p = 0.7$ rad/s and $E_f = 30.2$ kW/m could be formed by setting $\alpha = 0.00418$. Similarly, a TMA sea state with the same ω_p and with $E_f = 31.2$ kW/m could be formed by setting $\alpha = 0.0136$ at a water depth of 8 m.

4.5. Some issues with the irregular waves

In stationary waves, both $S(\omega_N)$ and ε_N , $N = 1, 2, \dots, 24$ are constants which mathematically represent a non-sinusoidal wave such as a higher order Stokes wave. However, in irregular waves caused by offshore

winds, storms and local winds, both $S(\omega_N)$ and ε_N , $N = 1, 2, \dots, 24$ are variables which effectively cause the sea state to change from time to time. In this paper, such issues are taken into consideration by using a large number of randomly selected sets of ε_N , $N = 1, 2, \dots, 24$ with the standard wave spectrums.

There are mainly two conditions specified for wave breaking in (Svendsen, 1983; Basco and Yamashita, 1987; Hansen, 1990; Svendsen, 1992; Svendsen, 2006) out of which, one is based on a limiting value of λ/h and the other on that of H/h . For all the sea states considered in JONSWAP spectrum and TMA spectrum, the 24 frequency components would not be broken in the free sea individually. However, inside the concentrator with the selected dimensions, those individual TMA wave components with $\omega < 0.7$ rad/s would be broken somewhere in the converging channel section. Note that in these cases of breaking, the condition on λ/h is satisfied but not that on H/h . Anyway, these broken frequency components are not included in the evaluations making the number of frequency components in the spectrum slightly less than 24. Any disturbances caused by such breaking is assumed to be negligible because $S(\omega_N)$; $\omega_N < 0.7$ rad/s is considerably small in the sea states considered.

In this context of representing irregular waves with wave spectrums, further attention is required on some other issues. Firstly, the aggregate wave height at some instants could sometimes be excessive depending on the prevailing phase distribution (ε_N , $N = 1, 2, \dots, 24$). In this case, the condition on H/h for breaking could be satisfied by the constructive addition of the instantaneous amplitudes of some wave components at certain time instants although it is not satisfied by the individual components. If the irregular wave is broken in this way, the buoy motion and the power absorption cannot be predicted using the foregoing analysis. Secondly, just by visual observation, it is not correct to conclude that there are no waves in the sea because it is also possible that the prevailing phase distribution of the irregular wave spectrum is such that the aggregate wave height is small due to cancellations. During such intervals, the buoy motion would be limited and the power absorption would be small. In practice, if such occurrences of wave breaking or no wave activities are found to be frequent in a certain site in the sea, WECs would be installed in a different site where unfavorable phase distributions are infrequent. In theory, this kind of breaking events or intervals with no wave activities are probabilistic and the percentage of phase distributions causing them could be evaluated using the laws of probability and statistics rigorously. However, in order to evaluate the corresponding effect on the estimated quantities such as the power absorption, the likelihood of those phase distributions or in other words, the percentage of time each phase distribution is in effect must be known.

Thirdly, it may also be the case that a certain irregular wave in the nearshore could be a resultant of addition of another few irregular wave fronts which could be reflections or oblique waves. Additionally, longer waves with higher phase velocities are known to overtake shorter waves (Thurman and Trulillo, 2001; Tissier et al., 2017). In these cases, nonlinear mixing and merging of wave components could be possible, effectively causing breaking or changing the sea state from time to time (Van Dongeren et al., 2007; Tissier et al., 2017). Any further consideration on this type of random mixing, merging or breaking is beyond the scope of this paper. Some more useful material on this regard are given in (Hansen, 1990; Svendsen, 2006; Kimmoun and Branger, 2007; Goda, 2010; Power et al., 2010). In practice, such areas in the sea could be avoided by referring the long term records of wave statistics (Barstow et al., 2009; Mørk et al., 2010). For example, wave reflections from the shore would be minimal in a wide sandy beach.

5. Semi-theoretical estimations

The power absorbed by both CDBD and CDBD-FWA point absorber type WECs could be evaluated by using the theory and the simulation results as described in Sections 3 and 4 for the standard models of wave

spectrums. In this case, once $z(\omega_N)$, $\dot{z}(\omega_N)$ and $P_{ave}(\omega_N)$ for all N are evaluated for a given sea state of a given wave spectrum, the spectral power absorption of the PTO is given by (Hughes, 1984; Pastor and Liu, 2014; Brask, 2015; Grüter, 2016)

$$P_{ave} = \sum_{N=1}^{24} P_{ave}(\omega_N) \quad (34)$$

Note that with this approach, the hydrodynamic coefficients m_a and R for each frequency component ω_N are found using the simulation results (Eriksson et al., 2005) making P_{ave} semi-theoretical. In addition, the mass of other moving parts such as the chain, the inertia of the gear-wheel including the shaft and the frictional and bearing losses are neglected. The motion equation is simplified by neglecting the dependence of F_w on z . The wave spectrum is reduced to a set of discrete frequencies. Considering all these limitations, hereinafter, P_{ave} found in this way would be referred to as estimates.

5.1. System resonance

Normally, there is a resonance frequency (ω_r) around which the WEC response $|H(j\omega)|$ is highest. However, ω_r is found to be much higher than the significant components of the sea wave spectrum with high energy density located around ω_p . In other words, around ω_r , the energy densities of the wave components are comparatively very small. Therefore, it is required to downshift ω_r of the WEC response to match ω_p of the wave spectrum to capture more power, if possible.

From Eq. (17), $|H(j\omega_r)|$ seems to be maximum when

$$\omega_r = \sqrt{C/A} = \sqrt{(K_h + K_s)/(M + m_a + 0.5m_1)} \quad (35)$$

However, note that m_a in this expression is frequency dependent and ω_r could only be found by trial and error unless a functional relationship is found for m_a . Yet, $B\omega$ term in Eq. (17) is not only frequency dependent but also B_{PTO} dependent. Therefore, Eq. (35) is not suitable for this purpose and the whole expression in Eq. (17) must be used to find the exact resonance frequency ω_r . This would be a more complicated trial and error procedure needing to substitute m_a and R values at each different frequency while adjusting B_{PTO} and m_1 for each sea state. In CDBD WEC, the procedure is somewhat simpler as m_1 is zero.

5.2. WEC response reshaping

Although $z(\omega_N)$ is a function of $|H(j\omega_N)|$, from Eq. (34), P_{ave} is given by

$$P_{ave} = \sum_{N=1}^{24} 0.5B_{PTO}|\dot{z}(\omega_N)|^2 \quad (36)$$

where $|\dot{z}(\omega_N)|^2$ is a function of $\omega_N^2|H(j\omega_N)|^2$ and is more frequency dependent. Hence, the contribution to P_{ave} could be high although $S(\omega_N)$ is small for high ω_N . Therefore, rather than down shifting ω_r , it would be more sensible to reshape the whole WEC response $H(j\omega_N)$; $N = 1, 2, \dots, 24$ with B_{PTO} and m_1 so that P_{ave} is maximal at a given sea state.

With response reshaping, a resonance with a narrow peak located close to the peak frequency of the wave spectrum could be observed for small values of B_{PTO} resulting in unusually high P_{ave} in both systems. Note that the accuracy and the reliability of the estimates provided by this semi-theoretical procedure is dependent on the condition that the buoy is always partly submerged in water. This condition can be checked by noting if the instantaneous down-most point of the buoy ($z_{dm}(t)$) is above water where

$$z_{dm}(t) = z(t) - d - \eta(t) \quad (37)$$

With small values of B_{PTO} in high sea states, it was indeed found that the buoy often leaves the water surface and makes in-air excursions with both WEC systems. In such cases, if the buoy takes too long to return to

the water, it takes too long to re-establish the phase relationship with the wave again. Hence, the estimated time average of the absorbed power ($P(\varepsilon_N)$) over a certain time window (T_W), would be less than the estimated spectral average (P_{ave}) given by (36) decreasing the reliability of the estimate where,

$$P(\varepsilon_N) = \frac{1}{T_W} \sum_{N=1}^{24} \left(\int_{T_W} P(t, \omega_N) dt \right) \quad (38)$$

Thus, too small a B_{PTO} increases the number of in-air excursions and decreases the estimation reliability (ER) of P_{ave} . Too high a B_{PTO} increases ER but decreases P_{ave} . This problem is severe in stronger waves with higher energy flux although the dependence on m_1 is relatively small in any case. Therefore, the procedure of maximizing P_{ave} by reshaping WEC response purely based on B_{PTO} and m_1 too is not satisfactory requiring some optimization. In optimal re-shaping of WEC response, it is attempted to keep P_{ave} as high as possible while ER is higher than at least 50% for each radius of the buoy and for each sea state. This convention implies that the average power absorption would be less than P_{ave} for (100 - ER) % of the time. In this paper, $z_{dm}(t)$ is evaluated at every 0.1 s over an interval of $T_W = 12.6$ s in order to estimate ER. If the buoy is found to be above the water surface at one or more of the 127 time instants, the estimate of P_{ave} is considered to be unreliable during that interval. The percentage of ER is evaluated by using 100 different such intervals. At each different interval, a different phase distribution (ε_N), $N = 1, 2, \dots, 24$; $K = 1, 2, \dots, 100$ is selected randomly.

The choice of ER > 50% for P_{ave} estimates is arbitrary and the real value is dependent on the type of the application. Similarly, small in-air excursions with $0 < z_{dm}(t) < \delta$ for small δ , could be neglected when evaluating ER. This is because the instantaneous value of F_w is really less than $F_{w \max}$ used in the frequency domain analysis of the system and the true number of in-air excursions could be less than that found with $z_{dm}(t) > 0$. Further, if the buoy is found to be above the water surface over just one or two consecutive time instants only, the time taken to return to the water and resynchronize with the wave would be small. Such occurrences too could be neglected when evaluating ER because the corresponding power loss would be small.

The responses of CDBD-FWA WEC systems with $r = 1$ m and 2 m were reshaped such that P_{ave} was optimal for all the sea states considered in JONSWAP and concentrated TMA wave spectrums. For example, illustrated in Fig. 2, is the optimally reshaped response of CDBD-FWA WEC system with $r = 2$ m and $M = 21343$ Kg, which was obtained by setting $B_{PTO} = 80$ kNs/m and $m_1 = 201879$ Kg for a JONSWAP sea state with $\omega_p = 0.9377$ rad/s, $\alpha = 0.0026$ and $E_{j0} = 4.65$ kW/m. In this case, $K_s = 5$ kN/m, $S = 8$ m and $MSL = 4$ m such that $d = 1.5$ m from Eq. (1). The corresponding value of ER with these settings is 89%. Illustrated in the same figure is the response of the CDBD system with the same parameters but $m_1 = 0$.

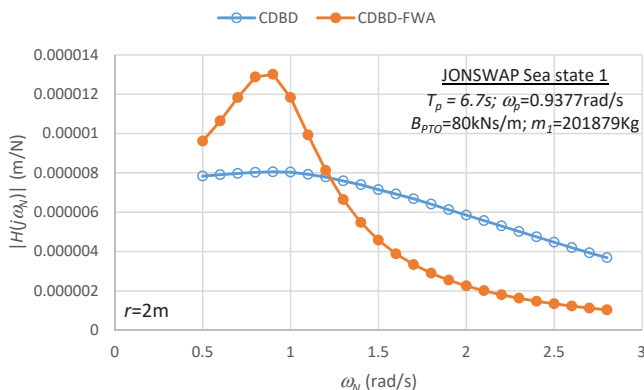


Fig. 2. Reshaped CDBD-FWA system response for a JONSWAP sea state.

6. Results

In this section, the upper bounds of P_{ave} and ER of CDBD WEC and CDBD-FWA WEC are presented for multiple sea states described by JONSWAP, TMA and concentrated TMA wave spectrums. The lower bounds are not shown because they are found to be always four times less than the corresponding upper bounds. This is because those terms due to the dynamic pressure (D_1) and the buoyancy (D_2) in Eq. (17) are equal in magnitude and P_{ave} is proportional to the square of $(D_1 + D_2)$ according to Eq. (20). In each case, the sea state parameters such as T_p , ω_p , H_{mo} , T_e and E_{j0} and the corresponding WEC system parameters such as r , m_1 and B_{PTO} are specified. The same values of m_1 , which were used for WEC response reshaping in the concentrated TMA spectrum, are used for TMA wave spectrum too because the differences were found to be small. Further, the same values of B_{PTO} are used for both CDBD-FWA and CDBD systems. In all cases, S and MSL have been set to 8 m and 4 m respectively but M has been adjusted such that d is always 1.5 m. For each type of WEC, results are presented for $r = 1$ m and 2 m and the corresponding K_s and M are shown in Table 1. Note for example that in the case of $r = 1$ m, $M = 6866$ Kg could be realized by setting the length of the buoy to 3 m and using a material with a density of 728.5 Kg/m³ such that 50% of the buoy is submerged when the water is still.

Once the wave spectrum ($S(\omega)$) of a given sea state of either JONSWAP, TMA or the concentrated TMA models has been formulated as described in Section 4.1, 4.2 and 4.3 respectively, the discrete version of it ($S(\omega_N)$) was formed according to Section 4.4. For each $S(\omega_N)$, z and $\dot{z}$ were evaluated using Eqs. (17) – (19) for all the frequency components in the spectrum with a certain random phase distribution, ε_N , $N = 1, 2, \dots, 24$. Then, using $\dot{z}$, the power absorption was estimated for each such sea state with and without response reshaping. In finding ER as described in Section 5.2, small in-air excursions were tolerated by setting $\delta = 0.25$ m for the reasons described in the same Section. The value of 0.25 m was set arbitrarily but ER was not found to be very sensitive for $0 < z_{dm}(t) < 0.25$ m anyway. For each sea state, the power absorptions with response reshaping given in this section are optimally high and the values of B_{PTO} and m_1 are to be set accordingly as mentioned in Section 3, in practice.

6.1. Power absorption in the deep sea

Two sea states of the deep sea were formed by using JONSWAP model as described in Section 4.1. The values of T_p and H_{mo} of them were selected to be similar to the typical values of the North Sea as shown in Table 2 (Eriksson et al., 2005; Pastor and Liu, 2014) so that comparisons of power absorptions could be made for JONSWAP waves. Such similarities could also be useful for comparing the power absorptions to those evaluated by using the time series of wave data in the North Sea (Eriksson et al., 2005) and by installing prototypes in the real North Sea (Eriksson et al., 2007; Sjökvist et al., 2014). The other relevant parameters of these two sea states together with the range of the frequency spectrums are also given in Table 2.

In this case, Sea state 2 with an energy flux of 30.2 kW/m is much more energetic than Sea state 1 with an energy flux of 4.65 kW/m as illustrated in Fig. 3. In both sea states, the normalized energy density $S_{JONSWAP}(\omega_N)$ approaches zero at lower and higher ends of the considered range of frequencies.

6.1.1. Velocity of the buoy and the short-term power absorption

The velocity of the buoy ($\dot{z}$) was found by the superposition of the

Table 1
Fixed parameters of both types of WECs.

Buoy radius, r	1	2
Spring stiffness, K_s (kNs/m)	5	5
Buoy mass, M (Kg)	6866	21343

Table 2

Parameters of the two JONSWAP sea states.

JONSWAP Sea state	Peak period T_p (s)	Peak frequency ω_p (rad/s)	α, γ	Range of 24 wave components (rad/s)	Mean significant height, H_{m0} (m)	Energy period T_e (s)	Energy flux E_{j0} (kW/m)
1	6.7	0.9378	0.0026, 3.3	0.5 – 2.8	1.25	6.09	4.65
2	8.81	0.7132	0.00418, 3.3	0.3 – 2.6	2.78	8.02	30.24

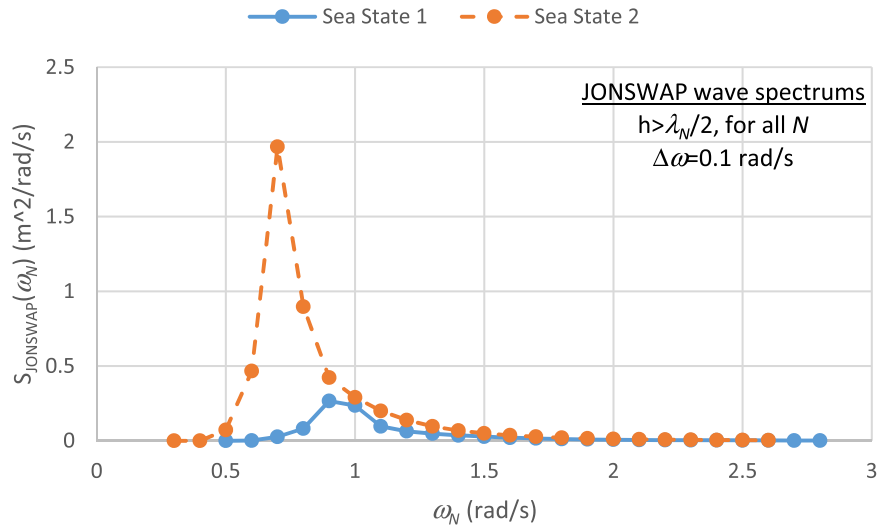


Fig. 3. Discretized JONSWAP wave spectra.

velocities due to the individual wave components. Illustrated in Fig. 4 is $\dot{z}$ for CDBD WEC with $r = 2$ m in Sea state 1 when the set of random phases $(\varepsilon_N, N = 1, 2, \dots, 24)$ are {225, 195, 270, 180, 45, 150, 285, 75, 165, 135, 60, 150, 30, 345, 330, 225, 150, 135, 240, 90, 15, 120, 270, 285} in degrees and B_{PTO} is 140 kNs/m. There were no in-air excursions with this particular phase distribution. The velocity ($\dot{z}$) due to a single wave component (0.9 rad/s or 1.0 rad/s) is sinusoidal but that due to the full spectrum (0.5 – 2.8 rad/s) is irregular. The time average of the power absorption $(P(\varepsilon_N))$ over 12.6 s is found to be 26.88 kW in this case but it would be different for a different set of $\varepsilon_N, N = 1, 2, \dots, 24$. Further, P_{ave} which is the spectral average of the same, is 30.32 kW and rather different from $P(\varepsilon_N)$. An average of $P(\varepsilon_N)$ taken over large number of different $\varepsilon_N, N = 1, 2, \dots, 24$ seemed to approach P_{ave} .

In JONSWAP Sea state 2, $P(\varepsilon_N)$ over 12.6 s of CDBD WEC with the

same parameters is illustrated in Fig. 5 for 100 different phase distributions, $(\varepsilon_N)_K, N = 1, 2, \dots, 24; K = 1, 2, \dots, 100$. It can be seen that $(P(\varepsilon_N)_K)$ could be as small as 19 kW and as high as 305 kW depending on K . The numerical average of these 100 time averages is 107.98 kW (dashed line in Fig. 5) which is somewhat close to the spectral average (P_{ave}) which is 113.79 kW. In practice, it is not known or not predictable currently how long a certain $\varepsilon_N, N = 1, 2, \dots, 24$ would be in effect in a given sea state with a constant T_p and H_{m0} . It might be the case that $\varepsilon_N, N = 1, 2, \dots, 24$ is constant until the sea state changes to a different one. Yet, when the same sea state occurs again sometime later, it is unlikely that $\varepsilon_N, N = 1, 2, \dots, 24$ is the same. Hence, a WEC system would generate different amounts of power from time to time even if the sea state is the same. In addition, the possibility that the buoy becomes airborne could be high for some phase distributions reducing $P(\varepsilon_N)$, as

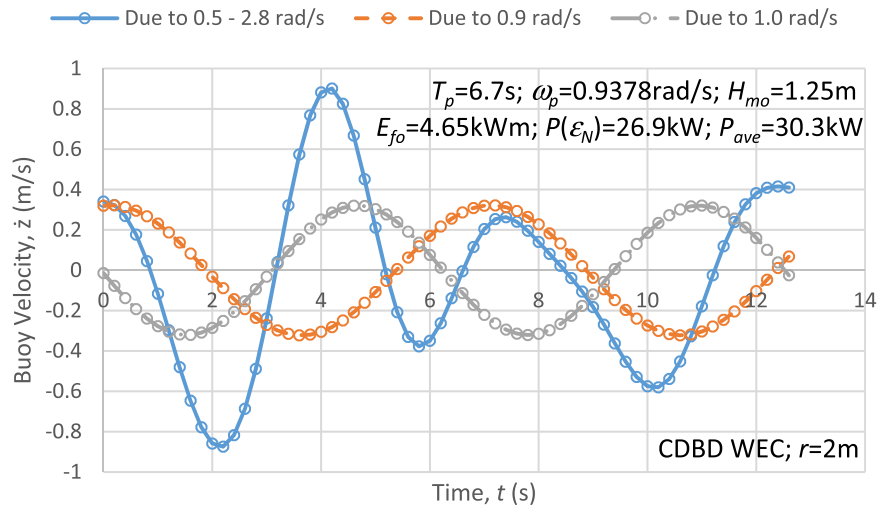


Fig. 4. Buoy velocity of CDBD WEC when $r = 2$ m in JONSWAP Sea state 1.

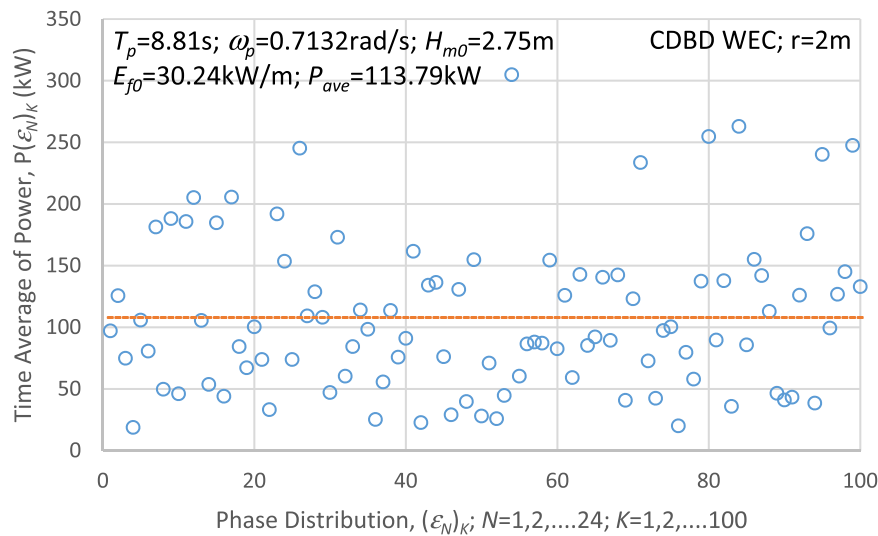


Fig. 5. $P(\varepsilon_N)_K$ of CDBD WEC for 100 phase distributions.

mentioned earlier. In this way, the absorbed power over even few days could be much smaller than the expected spectral average (P_{ave}).

6.1.2. The spectral average of power absorption and the reliability

The estimates of P_{ave} and ER of the two WECs with $r = 1$ m are given in Tables 3 and 4. In Sea states 1 and 2, response of the CDBD-FWA WEC was found to be optimal when B_{PTO} is 40 kNs/m and 80 kNs/m, and m_I is about 49MT and 82 MT respectively. Note in Table 4 that P_{ave} have been increased in CDBD-FWA WEC by about 24% and 16% in these two sea states. Although ER is 100% in Sea state 1 with both systems, it is lower in Sea state 2 with CDBD-FWA WEC. The obvious reason for this lower ER is that CDBD-FWA WEC with such high flywheel mass has a much higher inertia. It is difficult to prevent the buoy from leaving the water once this system has gained velocity and momentum.

In the point-absorber WEC mounted on an oil rig described in (Pastor and Liu, 2014), P_{ave} is about 15 kW and 40 kW in more or less the same two JONSWAP sea states respectively. With CDBD-FWA WEC, the upper bounds are about 8.4 kW and 22.2 kW respectively. The considerably higher power in the former case could be attributed to the conical shape used for the buoy although the radii of both types are 1 m. However, this tendency is changed when r is increased. In addition, the possibility of the buoy becoming airborne from time to time, seems to have been neglected in that case.

The estimates of P_{ave} and ER together with the other relevant parameters of the two WECs with $r = 2$ m are given in Tables 5 and 6. It can be seen that the improvement of power absorption possible with response reshaping is about 54% and 21% in Sea state 1 and 2 respectively. Note that the power absorption possible with CDBD-FWA WEC is as high as 114.1 kW when the energy flux is 30.2 kW/m but the required values of B_{PTO} and m_I too are much higher. The possibility of in-air excursions, particularly with CDBD-FWA, is also high resulting in somewhat low values of ER.

For comparison, P_{ave} of the point absorber WEC described in (Pastor and Liu, 2014), is about 33 kW and 85 kW in the same two JONSWAP

sea states respectively. However, in these sea states, P_{ave} of CDBD-FWA WEC is about 55 kW and 114 kW respectively. The fact that these values are upper bounds is compensated by the considerably high margins of 22 kW and 29 kW respectively. Note that these are somewhat higher in CDBD WEC too. The higher P_{ave} could be partly attributed to the lower draft of 1.5 m set in CDBD WECs. In contrast, a draft of 2 m has been set for the WEC described in (Pastor and Liu, 2014). Although, the number of in-air excursions is higher and ER is lower with the lower draft, this possibility is not considered in the other. For both WEC systems and for both radii, P_{ave} is illustrated together with the energy flux of each corresponding sea state in Fig. 6. Note that the difference in P_{ave} is small when $r = 1$ m and hence the corresponding data points are less distinguishable particularly in Sea state 1 with a low energy flux of 4.65 kW/m.

6.2. Power absorption in the nearshore

The nearshore wave spectrums were formulated using TMA model as described in Section 4.2 for a still water depth of $h = 8$ m. The parameters of four different sea states with $\omega_p = 0.7, 0.8, 0.9$ and 1.0 rad/s are listed in Table 7. In this case, ω_p , T_p and H_{m0} of Sea states 2 and 4 are approximately similar to those of the two JONSWAP sea states and therefore the power absorption of the WECs in the nearshore and in the deep sea could be compared in some way. Additionally, the energy flux of 31.17 kW/m in Sea state 4 is almost the same as that of Sea state 2 in JONSWAP model. The ranges of the frequency components used in the discretized spectrums are also listed in Table 7. Note that when ω_p is smaller, these ranges have been shifted towards lower frequencies so that the number of wave components in the spectrum are the same.

The corresponding TMA wave spectrums are illustrated in Fig. 7. Again, those spectrums with lower ω_p have been formed to have higher H_{m0} and $S_{TMA}(\omega_N)$ because the energy density of longer waves are usually higher. Note that in all four sea states, $S_{TMA}(\omega_N)$ is nearly zero at lower and higher ends of the considered range of frequencies.

The nearshore estimates of P_{ave} and ER of the two WECs with $r = 1$ m is given in Tables 8 and 9. In this case, response of the CDBD-FWA WEC was found to be optimal when B_{PTO} is 40, 40, 80 and 80 in kNs/m and m_I is about 22, 36, 38 and 52 in MT respectively for Sea states 1, 2, 3 and 4. Note that the advantage of CDBD-FWA over CDBD with respect to the percentage increase in P_{ave} is less in the nearshore than in the deep sea. On the positive side, ER is either 100% or almost 100% with both systems, except in Sea state 4. The estimates of P_{ave} in the nearshore is about 1–3 kW higher than that in the deep sea with both systems when considering those similar sea states and for the same values of B_{PTO} . Note

Table 3
 $r = 1$ m, CDBD WEC (with no WEC response reshaping) in JONSWAP waves.

JONSWAP Sea state	Energy flux E_{f0} (kW/m)	PTO damping B_{PTO} (kNs/m)	Absorbed power of CDBD WEC P_{ave} , (kW)	Estimation reliability with 0.25 m tolerance, ER (%)
1	4.65	40	6.8	100
2	30.24	80	19.1	96

Table 4 $r = 1$ m CDBD-FWA WEC (with WEC response reshaping) in JONSWAP waves.

JONSWAP Sea state	Energy flux E_{f0} (kW/m)	PTO damping B_{PTO} (kNs/m)	Flywheel mass m_I (Kg)	Absorbed power of CDBD-FWA WEC P_{ave} (kW)	Increase of P_{ave} w. r. t. CDBD WEC (%)	Estimation reliability with 0.25 m tolerance, ER (%)
1	4.65	40	49069	8.4	23.5	100
2	30.24	80	81782	22.2	16.2	74

Table 5 $r = 2$ m, CDBD WEC in JONSWAP waves.

JONSWAP Sea state	Energy flux E_{f0} (kW/m)	PTO damping B_{PTO} (kNs/m)	Absorbed power of CDBD WEC P_{ave} (kW)	Estimation reliability with 0.25 m tolerance ER, (%)
1	4.65	80	35.5	100
2	30.24	240	94.6	88

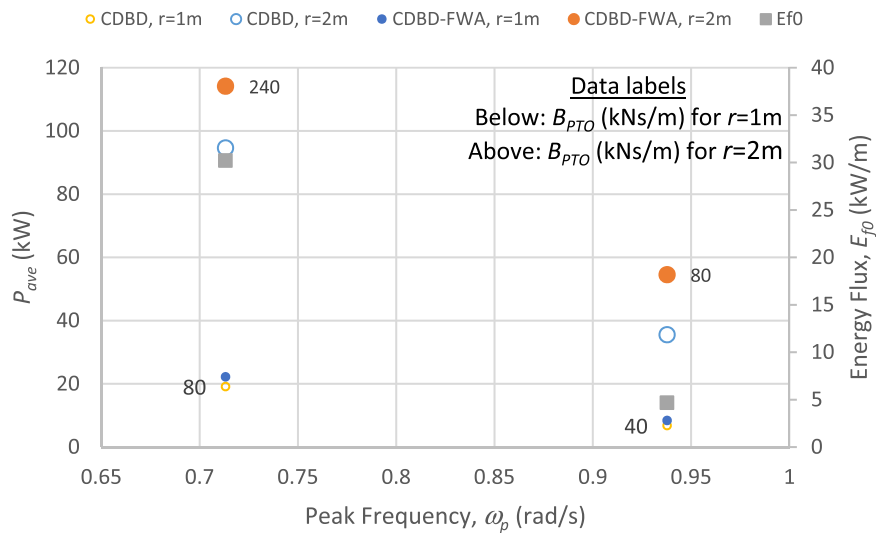
that if P_{ave} in a given sea state in the deep sea is monitored continuously while propagating towards the nearshore, P_{ave} in the nearshore would be lower due to the losses (Sinha et al., 2016). The increase of P_{ave} in similar sea states in the nearshore mentioned here could be attributed to the

differences in the TMA and JONSWAP wave spectrums. In TMA spectrum, more energy is distributed at higher frequencies and the power absorption is proportional not only to the energy density but also to the square of the frequency. However, ER, being a function of the phase distribution, is similar in both the nearshore and the deep sea in these two sea states.

In (Sinha et al., 2016), the absorbed power of a point-absorber type WEC, consisting of a buoy with a conical shape and a hydraulic PTO sub system, has been evaluated using theory and simulations for TMA wave states at the water depths of 79 m, 58 m and 20 m. Although the water depth considered in this paper is 8 m, T_p of 7.9 s and H_{m0} of 1.98 m of the sea state 3 specified in Table 7 are found to be very close to those of one of the sea states at the water depth of 20 m considered in (Sinha et al., 2016). For this sea state, P_{ave} of the WEC systems are listed in Table 10.

Table 6 $r = 2$ m CDBD-FWA WEC in JONSWAP waves.

JONSWAP Sea state	Energy flux E_{f0} (kW/m)	PTO damping B_{PTO} (kNs/m)	Flywheel mass m_I (Kg)	Absorbed power of CDBD-FWA WEC P_{ave} (kW)	Increase of P_{ave} w. r. t. CDBD WEC (%)	Estimation reliability with 0.25 m tolerance ER, (%)
1	4.65	80	201879	54.5	53.5	89
2	30.24	240	275289	114.1	20.6	75

**Fig. 6.** Power absorption of the two systems in the two JONSWAP sea states.**Table 7**

Parameters of TMA sea states.

TMA Sea state	Peak period T_p (s)	Peak frequency ω_p (rad/s)	Range of 24 wave components (rad/s)	Mean significant height, H_{m0} (m)	Energy period T_e (s)	Energy flux E_{f1} (kW/m)
1	6.28	1	0.5 – 2.8	0.89	5.01	2.25
2	6.98	0.9	0.5 – 2.8	1.43	5.6	6.42
3	7.85	0.8	0.4 – 2.7	1.98	6.3	13.6
4	8.98	0.7	0.3 – 2.6	2.86	7.24	31.17

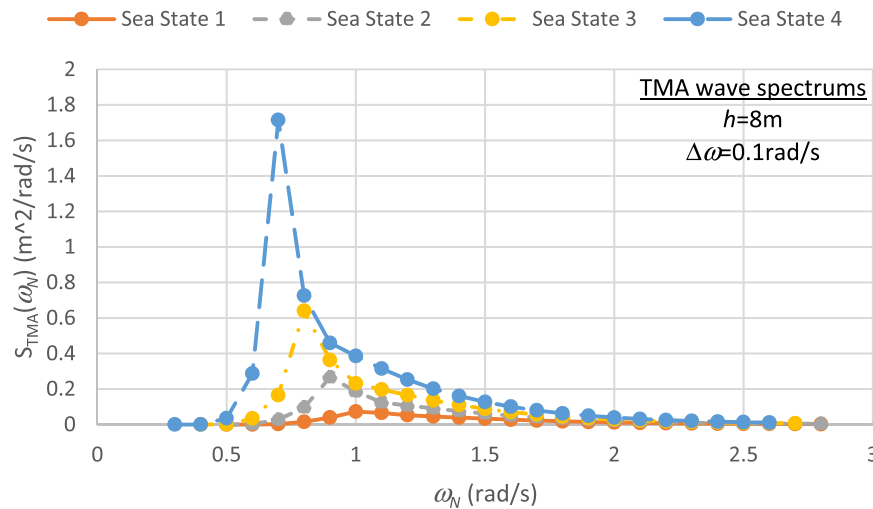


Fig. 7. Discretized TMA wave spectrums.

Table 8

 $r = 1$ m, CDBD WEC in TMA waves.

TMA Sea state	Energy flux (kW/m)	PTO damping B_{PTO} (kNs/m)	Absorbed power of CDBD WEC P_{ave} (kW)	Estimation reliability with 0.25 m tolerance ER, (%)
1	2.25	40	4	100
2	6.42	40	9.5	100
3	13.6	80	10.7	100
4	31.17	80	21.2	94

Note that the energy fluxes in these two cases, which are 14.14 and 13.6 in kW/m too are close enough for comparison.

In all the three systems compared in Table 10, the best B_{PTO} at which the P_{ave} is highest have been used. Therefore, m_1 in CDBD-FWA has been made different from that shown in Table 9. In the case of CDBD WEC, P_{ave} is obviously lower as the system is not matched to the wave spectrum of this sea state. Although P_{ave} is highest in CDBD-FWA, which is 28.7 kW, ER is decreased to just more than 50% due to the change in m_1 . Note further that in CDBD systems, the values of P_{ave} are upper bounds. On the other hand, it is not clear if the possibility of in-air excursions has been considered in the case of the point absorber WEC with the hydraulic PTO system. Hence, the performance of CDBD-FWA WEC and the WEC with the hydraulic PTO could considered to be more or less the same in this nearshore sea state.

The same parameters of interest of the two CDBD systems for a buoy with $r = 2$ m are listed in Tables 11 and 12. Again, the P_{ave} of CDBD-FWA WEC is higher but with percentages ranging from 6.8 to 12.6 only. With both systems ER is either 100% or very close to 100% in the first three sea states. In Sea state 4, response of CDBD-FWA system could not be adjusted to obtain any better value of ER even by allowing P_{ave} to decrease. Sea state 4 in the nearshore and Sea state 2 in the deep sea have similar T_p and H_{mo} and have almost the same energy flux. However,

Table 9

 $r = 1$ m CDBD-FWA WEC in TMA waves.

TMA Sea state	Energy flux (kW/m)	PTO damping B_{PTO} (kNs/m)	Flywheel mass m_1 (Kg)	Absorbed power of CDBD-FWA WEC P_{ave} (kW)	Increase of P_{ave} w. r. t. CDBD WEC (%)	Estimation reliability with 0.25 m tolerance ER, (%)
1	2.25	40	21808	4.3	7.5	100
2	6.42	40	35829	10.6	11.6	100
3	13.6	80	38165	11.4	6.5	99
4	31.17	80	52142	23.1	9	79

the P_{ave} possible in the nearshore are 11 kW and 3 kW higher with CDBD WEC and CDBD-FWA WEC respectively at the same B_{PTO} . This is again due to the difference in JONSWAP and TMA wave spectrums. Hence, the nearshore is more favorable for energy harnessing using point absorbers, given the same energy flux.

Illustrated in Fig. 8 are P_{ave} of CDBD and CDBD-FWA WECS for both radii. The energy flux of each sea state at this depth of 8 m is also illustrated. When $r = 1$ m, power absorption possible with both systems are similar and hence the data points are overlapped except at Sea state 4. The tendency of course is that the power absorption is increasing when the energy flux is increasing but the estimation reliability (ER) starts decreasing from 100%.

Table 10

Comparison to a point absorber with a hydraulic PTO.

	h (m)	T_p (s)	H_{mo} (m)	E_{f1} (kW/m)	P_{ave} (kW)	ER (%)
Hydraulic PTO	20	8	2	14.14	25	-
CDBD	8	7.9	1.98	13.6	20.3	95
CDBD-FWA	8	7.9	1.98	13.6	28.7	56

Table 11

 $r = 2$ m, CDBD WEC in TMA waves.

TMA Sea state	Energy flux (kW/m)	PTO damping B_{PTO} (kNs/m)	Absorbed power of CDBD WEC P_{ave} (kW)	Estimation reliability with 0.25 m tolerance ER, (%)
1	2.25	80	23.9	100
2	6.42	180	36.2	100
3	13.6	240	54.2	100
4	31.17	240	105.7	82

Table 12

$r = 2$ m CDBD-FWA WEC in TMA waves.

TMA Sea state	Energy flux (kW/m)	PTO damping B_{PTO} (kNs/m)	Flywheel mass m_f (Kg)	Absorbed power of CDBD-FWA WEC P_{ave} (kW)	Increase of P_{ave} w. r. t. CDBD WEC (%)	Estimation reliability with 0.25 m tolerance ER, (%)
1	2.25	80	68822	26.9	12.6	100
2	6.42	180	91737	38.8	7.2	100
3	13.6	240	132088	57.9	6.8	98
4	31.17	240	193477	117.3	11	56

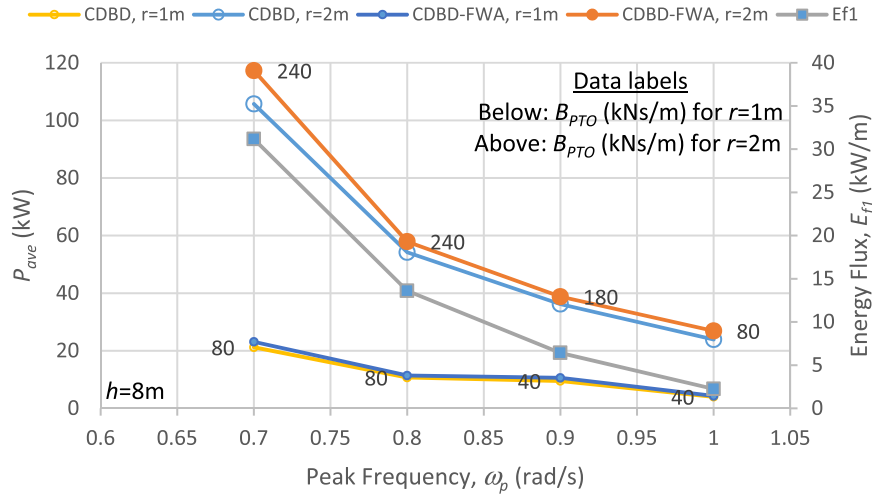


Fig. 8. Power absorption of the two systems in TMA sea states.

6.3. Power absorption in the concentrated nearshore waves

In concentrating nearshore waves, the dimensions of the concentrator and the placement must be determined carefully as mentioned in Section 4.3. It has been shown in (Jagath Kumara, 2022), that the widths and the lengths of the three sections of the concentrator must be in the order of a wave length. A concentrator with dimensions in the order of a few meters is not capable of passing the most common sea waves with periods ranging from 5 to 10 s. In the case of irregular waves such as those represented by TMA model, the wave length of ω_p could be considered as a reference for setting the dimensions. However, lowest frequencies of the wave spectrum would still be rejected by the concentrator. Out of the propagating waves, again the lowest frequencies would be broken inside. This is not troublesome with the TMA wave spectrum because it is very narrow for $\omega < \omega_p$ with sharply decreasing energy density. For example, as illustrated in Fig. 7, $S_{TMA}(\omega_N)$ is almost zero just 0.2 rad/s below ω_p in all the four sea states. Because, the response of this kind of a concentrator is similar to that of a high pass filter, the dimensions could be set such that those wave frequencies which are higher than a certain ω_b would be passed and concentrated. In this case, ω_b is the lowest unbroken wave frequency below ω_p , at which $S_{TMA}(\omega_b)$ is not negligible. The placement options of such a concentrator has been analyzed in (Jagath Kumara, 2022) and shown that a place closest to the shore would be more beneficial as long as the entrance is in the shoaling region of waves. In this case, the shoaling region must be determined by considering the most common wave frequencies in a given coast or by visual observation. Additionally, when placing point-absorber type WECS for energy harnessing, the water depth (h) must be higher than the possible downward movement of the buoy below SWL.

The lengths, widths and the water depths of each section of the concentrator used for the analysis of CDBD WECs in this paper are listed in Table 13. The remaining part of this paragraph on the various

quantities in Table 13 is based on the previous work of the same author described in (Jagath Kumara, 2022). In a plane beach with a slope of $h_x = 1/60$, none of the wave components ranging from 0.5 rad/s to 2.8 rad/s are subject to breaking 480 m away from the shore at 8 m depth in the free sea. Hence, the inlet of the concentrator has been placed at this area. The length of each section has been set to 60 m which is approximately the average wavelength of the probable peak frequencies in the nearshore at this depth. The widths given in Table 13 have been set such that ω_b is 0.8 rad/s. The implication is that all those wave components below 0.8 rad/s would be either filtered or broken inside. Hence, only the first three TMA sea states in Table 7 could be useful for forming the concentrated wave spectrums because even ω_p is broken in Sea state 4. If ω_b are to be decreased, the width of the uniform channel (w_2) needs to be increased. Then, in turn, the concentration gain for the sea states with higher ω_p would be decreased because the gain is equal to w_1/w_2 where w_1 is the width of the inlet of the converging channel. Therefore, ω_b could be set properly only after referring the wave statistics in a given coast. According to the widths specified in Table 13, the gain in concentration would be $100/38$ which is about 2.6. Therefore, Mid-channel energy flux (E_{fmid}) of those unfiltered and unbroken wave components would be increased by a factor of 2.6 neglecting losses. The width of the exit of the diverging channel (w_3) has been set to 75 m. However, it was found later that it would have been more effective if it were set to be wider.

Table 13

Set of dimensions of the concentrator.

Section	Converging Channel		Uniform Channel	Diverging Channel	
Location	Inlet	Exit	Mid-channel	Inlet	Exit
Distance from shore, D (m)	480	420	390	360	300
Width, w (m)	100	38	38	38	75
Still water depth, h (m)	8	7	6.5	6	5

The parameters such as the wave number, wavelength, phase and group velocities, wave height, energy density and the energy flux of each wave component in the three sea states were evaluated at each point of interest in this concentrator as described in (Jagath Kumara, 2022). Listed in Table 14 are the energy fluxes of the concentrated TMA waves together with those of the TMA waves in the free sea near the inlet of the concentrator.

Illustrated in Fig. 9 are TMA wave spectrums before and after the concentration in Sea state 1 where $\omega_p = 1$ rad/s. Due to the concentration, the energy flux is increased from 2.25 kW/m to 5.88 kW/m. The spectrum of JONSWAP waves with the same ω_p and the same energy flux (5.88 kW/m) but with a small offset of 0.09 m in H_{m0} is also shown. Note that the spectrum of the concentrated TMA waves has a certain tendency to match the shape of JONSWAP spectrum. More energy is centered around ω_p in these two spectrums than that in TMA spectrum. Therefore, waves which are somewhat similar to the offshore waves, could be realized in the nearshore by concentration.

6.3.1. In-air excursions of the buoy

The possibility of the buoy crossing the air-water boundary is illustrated in this sub section because it is more pronounced in concentrated TMA waves due to the increases in heights of wave components. The buoy becomes airborne when the down-most point ($z_{dm}(t)$) is above the water surface in which case, $z_{dm}(t) > 0$ in Eq. (37). This possibility is different for different sea states and for different phase distributions. Illustrated in Fig. 10 is $z_{dm}(t)$ of a buoy with $r = 1$ m, which is connected to CDBD-FWA WEC system, for two phase distributions, $(\epsilon_N)_K$, $N = 1, 2, \dots, 24$; $K = 1, 2$ in Sea state 3. The sea state information, buoy mass, flywheel mass and the upper bound (UB) as well as the lower bound (LB) of absorbed power at B_{PTO} of 60 kN/m are also included in this figure. Note that $z_{dm}(t)$ becomes positive and the buoy becomes airborne on one occasion for Phase distribution 2 ($K = 2$). During this time, motion of the buoy is independent of the wave hydrodynamics and the motion equation in Eq. (8) is unreliable for power estimation. The occurrence of this event is marked with an arrow in Fig. 10. The absorbed power given by Eq. (38) over a finite time interval of $T_W = 12.6$ s is less than the possible value for Phase distribution 2. Note however that ER is evaluated with a tolerance of 0.25 m for $z_{dm}(t)$ as described in Section 5.2 and in the beginning of Section 6.

Normally, ER is less than 100% when H_{m0} and E_{fmid} are higher and ϵ_N , $N = 1, 2, \dots, 24$ is such that more frequency components of z add constructively. Therefore, low values of ER also partially indicate that there is a possibility of wave breaking. This is only a partial indication because z and η have a phase difference of θ where θ is given by Eq. (18). When components of z add constructively, those of η may not necessarily or vice versa. The issue of $ER < 100\%$ or wave breaking should not be a big concern anyway because the buoy would still heave and generate power but not according to the foregoing analysis. Low values of ER or too many breaking events would only make the power absorption less than the spectral average.

6.3.2. The power absorption and ER

For concentrated TMA waves, P_{ave} and ER are listed in Tables 15 and 16 for CDBD WEC and CDBD FWA WEC respectively when $r = 1$ m. The

energy flux before concentration (E_{f1}) and after concentration (E_{fmid}) are given in addition to the other relevant parameters. The increases in P_{ave} with response reshaping (CDBD-FWA WEC) are about 6% – 15% as shown in Table 16. Further, P_{ave} of both WECs are about 7 kW, 21 kW and 25 kW higher than those before concentration listed in Tables 8 and 9 in the same corresponding TMA sea states. In other words, P_{ave} possible with wave concentration is almost three times for both WECs and for all the three sea states. Note that this factor of three is close to the concentration gain of 2.6. Further, E_{fmid} in Sea state 3 (after concentration) and E_{f1} in Sea state 4 (before concentration) are more or less similar but P_{ave} possible in the former sea state is about 13 kW higher, which is about 62%. This increase could be attributed to the difference in the values of H_{m0} which are 3.53 m and 2.86 m respectively. The point absorbers are known to be more efficient when the wave heights are high. However, note that ER is almost 100% before concentration in these three sea states with both WECs but it is less after concentration in Sea state 2 and 3.

The same set of results and parameters are listed for $r = 2$ m in Tables 17 and 18. In this case too, about 8% - 15% increases in P_{ave} is possible with WEC response reshaping. The increases possible with wave concentration are very high when compared to those with no concentration (Tables 11 and 12). The corresponding increases in the three sea states are approximately 50 kW, 80 kW and 120 kW and are about three times as a factor. Although E_{fmid} in Sea state 3 and E_{f1} in Sea state 4 are similar, P_{ave} is about 69 kW higher in the former Sea state. Note that both with or without response reshaping, close to 180 kW of power could be harnessed in Sea state 3 by concentrating just 13.6 kW/m TMA waves in the nearshore. However, ER is lower in concentrated waves, which is almost 100% with no concentration.

Illustrated in Fig. 11 are P_{ave} of both WECs with both radii. The energy flux of each sea state before the concentration at a depth of 8 m is also shown. When the energy flux is increasing P_{ave} is increasing but the estimation reliability (ER) is decreasing. Interestingly, P_{ave} possible with wave concentration is three times higher for both WECs, both radii and for all the three sea states, which is close to the concentration gain of 2.6.

In the preceding analysis, in both WECs with both radii, the spring stiffness was set to $K_s = 5$ kN/m so that the buoy draft $d = 1.5$ m in still water as specified in Table 1. If K_s is varied, the resulting variation of d is found to be small particularly for $r = 2$ m because $Mg \gg K_s L$ in Eq. (1). Therefore, the same set of hydrodynamic coefficients (Eriksson et al., 2005) could be used for finding the effect of K_s on the power absorption. In this way, it was found that P_{ave} is increased only by a small amount when K_s is increased up to 10 kN/m in WECs with $r = 1$ m but the corresponding ER is decreased by a small amount too. Then, P_{ave} is increased by about 10 kW when K_s is increased up to 40 kN/m in CDBD WEC with $r = 2$ m and the corresponding ER is decreased by about 10%. In CDBD-FWA WEC with $r = 2$ m, this increase in P_{ave} and the decrease in ER would be about 5 kW and 5% respectively. In all of these cases, the increases in P_{ave} mentioned here are more or less constant over the three sea states. However, there is a small variation in ER values and mentioned here are the averages.

Table 14
Concentrated TMA wave spectrums.

Sea state	Peak frequency, ω_p (rad/s)	TMA wave spectrums in the free sea at $h = 8$ m		Concentrated TMA wave spectrums in Mid-channel at $h = 6.5$ m		
		Mean significant height, H_{m0} (m)	Funnel inlet energy flux, E_{f1} (kW/m)	Mean significant height, H_{m0} (m)	Mid-channel energy flux, E_{fmid} (kW/m)	Range of wave components (rad/s)
1	1	0.89	2.25	1.55	5.88	0.8 – 2.8
2	0.9	1.43	6.42	2.52	16.34	0.8 – 2.8
3	0.8	1.98	13.6	3.53	32	0.8 – 2.8

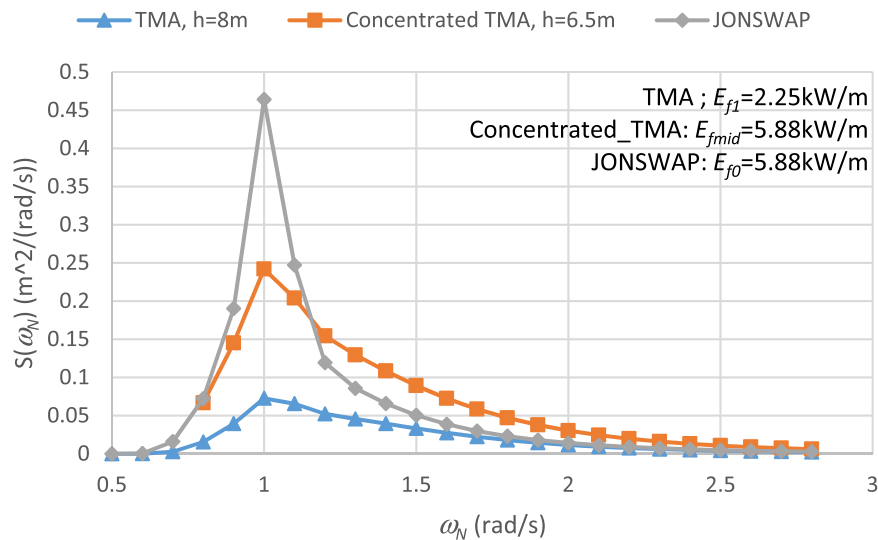


Fig. 9. Spectrum of the concentrated TMA Sea state 1 with comparisons.

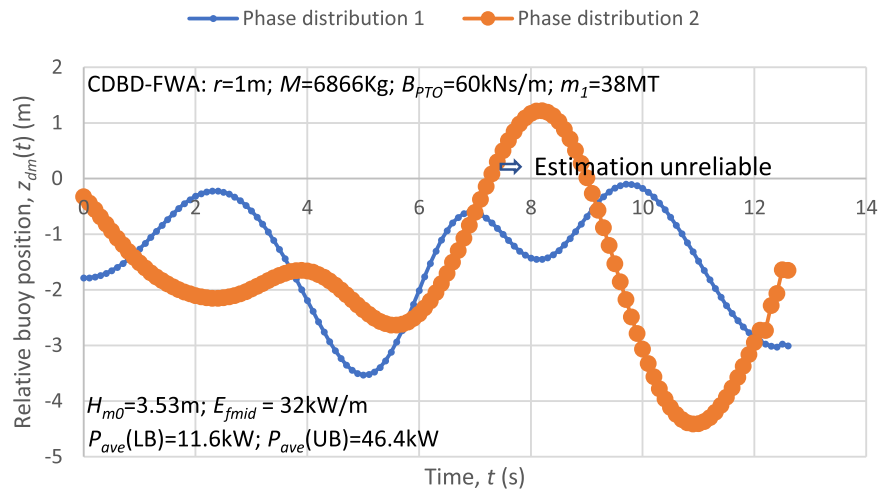


Fig. 10. $z_{dm}(t)$ in Sea state 3 of concentrated nearshore waves.

Table 15
 $r = 1$ m, CDBD WEC in concentrated TMA waves.

Nearshore Sea state	Nearshore energy flux, E_{f1} (kW/m)	Mid-channel energy flux, E_{fm1d} (kW/m)	PTO damping B_{PTO} (kNs/m)	Absorbed power of CDBD WEC P_{ave} (kW)	Estimation reliability with 0.25 m tolerance ER, (%)
1	2.25	5.88	40	11.7	100
2	6.42	16.34	40	29	87
3	13.6	32	80	34	64

Table 17
 $r = 2$ m, CDBD WEC in concentrated TMA waves.

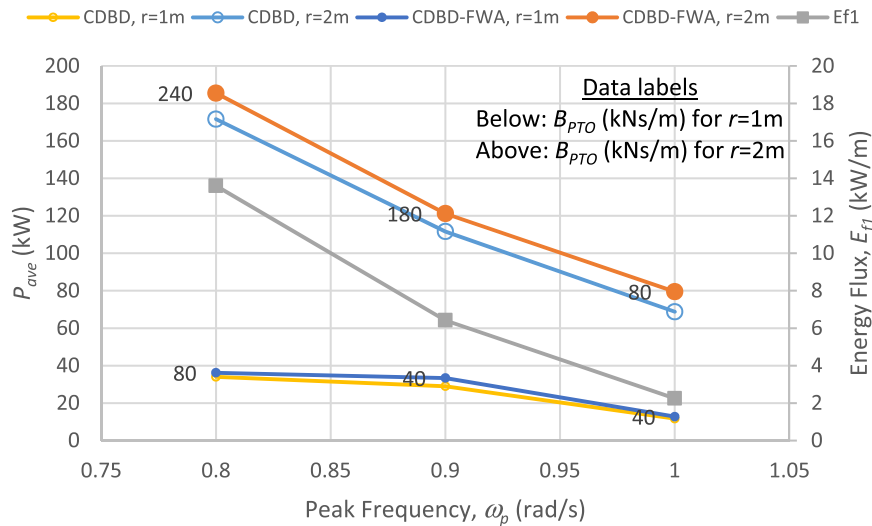
Nearshore Sea state	Nearshore energy flux, E_{f1} (kW/m)	Mid-channel energy flux, E_{fm1d} (kW/m)	PTO damping B_{PTO} (kNs/m)	Absorbed power of CDBD WEC P_{ave} (kW)	Estimation reliability with 0.25 m tolerance ER, (%)
1	2.25	5.88	80	68.8	98
2	6.42	16.34	180	111.6	86
3	13.6	32	240	171.6	62

Table 16
 $r = 1$ m CDBD-FWA WEC in concentrated TMA waves.

Nearshore Sea state	Nearshore energy flux, E_{f1} (kW/m)	Mid-channel energy flux, E_{fm1d} (kW/m)	PTO damping B_{PTO} (kNs/m)	Flywheel mass m_1 (Kg)	Absorbed power of CDBD-FWA WEC P_{ave} (kW)	Increase of P_{ave} w. r. t. CDBD WEC (%)	Estimation reliability with 0.25 m tolerance ER, (%)
1	2.25	5.88	40	21808	12.8	9.4	97
2	6.42	16.34	40	35829	33.4	15.2	67
3	13.6	32	80	38165	36.2	6.4	54

Table 18 $r = 2$ m CDBD-FWA WEC in concentrated TMA waves.

Nearshore Sea state	Nearshore energy flux, E_{f1} (kW/m)	Mid-channel energy flux, E_{fmid} (kW/m)	PTO damping B_{PTO} (kNs/m)	Flywheel mass m_f (Kg)	Absorbed power of CDBD-FWA WEC P_{ave} (kW)	Increase of P_{ave} w. r. t. CDBD WEC (%)	Estimation reliability with 0.25 m tolerance ER, (%)
1	2.25	5.88	80	68822	79.5	15.6	84
2	6.42	16.34	180	91737	121.2	8.6	69
3	13.6	32	240	132088	185.5	8.1	53

**Fig. 11.** Power absorption of the two systems in the concentrated TMA sea states.

6.4. The possibility of the buoy submersion

It is also possible that the buoy become submerged in water when the downward excursion is too much. If the waves are monochromatic, the probability of this event is the same as that of in-air excursions. However, in irregular waves such as in JONSWAP or TMA waves, $\eta(t)$ is not symmetrical around MSL. Therefore, even if the buoy become airborne due to a wave peak, it may not become submerged due to the next wave trough, and vice versa. Anyway, similarly to the case with in-air excursions, the motion equation in Eq. (8) is not applicable and the estimations are not reliable if the buoy is completely submerged. Note that the occurrence of any one of these events just once during a given time interval (t_W) is adequate to make the estimate of the time average of power $P(\epsilon_N)$ inaccurate. In this paper, such a time interval is excluded and counted when evaluating ER.

For all the three wave spectrums and the corresponding sea states, ER was evaluated by considering the possibility of the buoy becoming either airborne or submerged first in a given time interval, with the same tolerance of $\delta = 0.25$ m. For both CDBD and CDBD-FWA systems with both radii and with the same system parameters as before, $ER > 50\%$ in all the sea states considered in both JONSWAP and TMA waves. However, these values of ER are only somewhat less than the previous values obtained by considering the possibility of an in-air-excursion alone. This is also the case with the Sea state 1 and 2 of concentrated TMA waves. However, in Sea state 3, in which the energy flux is highest, ER is slightly less than 50% for CDBD-FWA system with both radii. Again, they are only slightly less than the previous values. Note that the values of spectral power estimates (P_{ave}) illustrated in Sections 6.1, 6.2 and 6.3 are not affected but their reliability is given by ER. In other words, $P(\epsilon_N)$ could be less than P_{ave} .

6.5. Overall comparisons

A quite a number of publications have been made on the analysis of the power absorption of point absorber type WECs. However, in most cases, not only the underlying system parameters such as r , M , B_{PTO} and K_s but also the wave model and the corresponding sea state parameters are different. Listed in Table 19 are the power absorptions of some of these WECs and those of some versions described in this paper for comparison. Note that only in a few cases, these parameters are found to be similar or close. The highest difference is observed in the case of the buoy mass (M) for a given radius. The power absorption has a nonlinear dependence on M according to Eqs. (17) and (36) and hence it is not really possible to make precise comparisons just by scaling. However, it is evident from these two equations that higher the total mass ($M + m_a + m_f$) higher the power absorption for a given r , K_s and B_{PTO} . Vaguely, the power absorption is initially increasing with M and then starts decreasing again for very large values of M . Note that the power absorption of $r = 2$ m WEC described in this paper (row 2) is almost seven times higher than that of the WEC described in (Potapenko et al. 2021) (row 4) in deep sea with more or less similar H_{mo} and T_e and the mass ratio is about seven times too. Further, a similar tendency can be observed in (Thomas et al., 2018) (row 6) in which case the mass ratio is about 4.3 times but however H_{mo} is about 1.5 times in a somewhat different wave spectrum. In the nearshore described by TMA spectrum, power absorption given in (Sinha et al., 2016) (row 10) is more or less similar in a similar sea state as mentioned earlier. Therefore, the indication is that the power absorptions of different versions in Table 19 are more or less within the same range. However, if the nearshore waves with an energy flux of 13.6 kW/m are concentrated about 2.6 times as described in this paper, nearly 185 kW could be harnessed with an $r = 2$ m WEC. In this case, no similar work could be found for comparison.

Table 19
Summary of results and comparisons to different point absorber designs.

	Sea spectrum	T_p (s)	T_e (s)	H_{mo} (m)	Energy Flux (kW/m)	Mass (M) Kg	Radius (r) (m)	Power absorption (kW)
[This paper]	JONSWAP	6.7	6.09	1.25	4.65	21343	2	54.5
[This paper]	JONSWAP	8.81	8.02	2.78	30.2	21343	2	114.1
(Mutombo and Numbi, 2022)	Pierson-Moskowitz	8.8	-	3.6	-	-	2	30
(Potapenko et al., 2021)	Real sea data	-	9	2.75	-	3000	2	16
(Tan et al., 2021)	JONSWAP	4.4	-	1.2	≈ 2	-	2.07	5
(Thomas et al., 2018)	Brettschneider	-	7.5	1.75	-	5000	2.5	5.3
(Pastor and Liu, 2014)	JONSWAP	8.81	-	2.75	-	-	2	85
[This paper]	TMA (nearshore)	7.9	6.3	1.98	13.6	6866	1	28.7*
[This paper]	TMA (nearshore)	8.98	7.24	2.86	31.17	21343	2	117.3
(Sinha et al., 2016)	TMA (nearshore)	8	-	2	14.14	-	-	25
[This paper]	Concentrated TMA (nearshore)	7.9	6.3	1.98	13.6	6866	1	36
[This paper]	Concentrated TMA (nearshore)	7.9	6.3	1.98	13.6	21343	2	185.5

* at a lower B_{PTO} than that in Table 9 for comparison.

6.6. Array implementations

The power absorption and the reliability of single WEC units have been quantified semi-theoretically in JONSWAP, TMA and concentrated TMA waves so far in Section 6. The highest power absorption possible with one unit of $r = 2$ m, CDBD-FWA WEC is found to be 114, 117 and 185 in kW respectively in these wave spectrums when the energy flux in the vicinity is about 30 kW/m. These values are somewhat lower with $r = 2$ m CDBD WEC. Therefore, if any higher power is required to be harnessed, multiple WEC units should be placed in the longshore direction to form an in-line array with enough spacing between them. The spacing among elements is the main parameter affecting the functionality of such an array. Due to reflections and deflections off multiple buoys, the incident wave would be modified (Engström et al., 2013; Stratigaki et al., 2014; Grüter, 2016). The wave spectrum and the phase distribution would be different from the expected due to the constructive or destructive additions of frequency components. One option is to space them such that the resultant wave is enhanced by the waves originated from nearby WEC units. However, this would be nearly impractical unless a few wave components around ω_p is selected for the purpose. Besides, the number of elements may be too many in an array and the distances travelled by those components with the same frequency would be different making it difficult to keep them at the same phase at a certain point. This approach could only be simplified to some extent by neglecting the effects from distant WEC units.

The best option would be to leave enough spacing to minimize the diffraction caused by the other WEC units. For example, over a 100 m crest width, the number of $r = 1$ m and $r = 2$ m units in an array would be about 16 and 6 if a spacing of about $4r$ and $6r$ is allowed respectively. These spacings are rather arbitrary but more research is still underway on this topic (Engström et al., 2013; Stratigaki et al., 2014; Murai et al., 2021). However, importantly, the corresponding number of WEC units could be decreased to six and three respectively in a wave concentrator with dimensions specified in Table 13. In this way, over the 38 m width of the uniform channel, the edge-to-edge spacing would be 3.7 m while

the center-to-center spacing would be 5.7 m with $r = 1$ m WECs. The same spacings would be 6.5 m and 10.5 m with $r = 2$ m WECs. Schematics of these arrays are illustrated in Fig. 12. If it is assumed that each WEC unit in the array functions independently from one another, the total power absorption would be given by

$$P_T = \Gamma P_{ave} \quad (39)$$

where Γ is the number of array elements which are identical. This condition would be rather easily satisfied if the inter-elemental spacing is high enough for low frequency waves such as sea waves (McIver and Bennett, 1993; Eriksson, 2005). Supposing that this is the case with the arrays specified in Fig. 12, the total power absorptions possible in concentrated TMA waves are illustrated in Fig. 13 for both WECs. Note that the parameters of the elements of these arrays are identical to those of which have been considered so far in Section 6. Both E_{fl} and the corresponding E_{fmid} of the three sea states are also shown in the same figure. Obviously, the 3-element array with $r = 2$ m units is much better than the 6-element array with $r = 1$ m units. When the energy flux of the incoming wave is just 13.6 kW/m (Sea state 3), 556 kW of P_T is possible by wave concentration and by placing a 3-element array with $r = 2$ m CDBD-FWA WEC units. Because P_{ave} of single CDBD WEC units are somewhat lower, P_T of those arrays formed by using CDBD WECs are always lower. Note for example that when $r = 2$ m, P_T of CDBD arrays are about 40–50 kW less than those of CDBD-FWA arrays in these three sea states. The accuracy of the power absorptions predicted in Fig. 13 by using Eq. (39) could be confirmed according to the results of (Murai et al., 2021). In this case, each single element of a 3-element in-line array has been shown to absorb 99.9% of the power that would be absorbed by a single isolated element. Further, by controlling the diffraction as mentioned earlier in this section, up to 113% increase in power absorption per each element has been possible (Murai et al., 2021).

According to P_T illustrated in Fig. 13, only a certain proportion of the available energy of the oncoming wave is absorbed by in-line arrays. Except for the radiation loss, the down wave which propagates towards the shore contains the remaining energy. Therefore, it is possible to

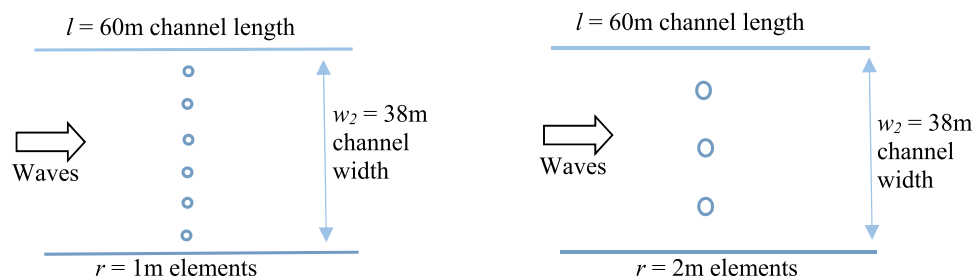


Fig. 12. Structure of the longshore, in-line arrays inside the concentrator.

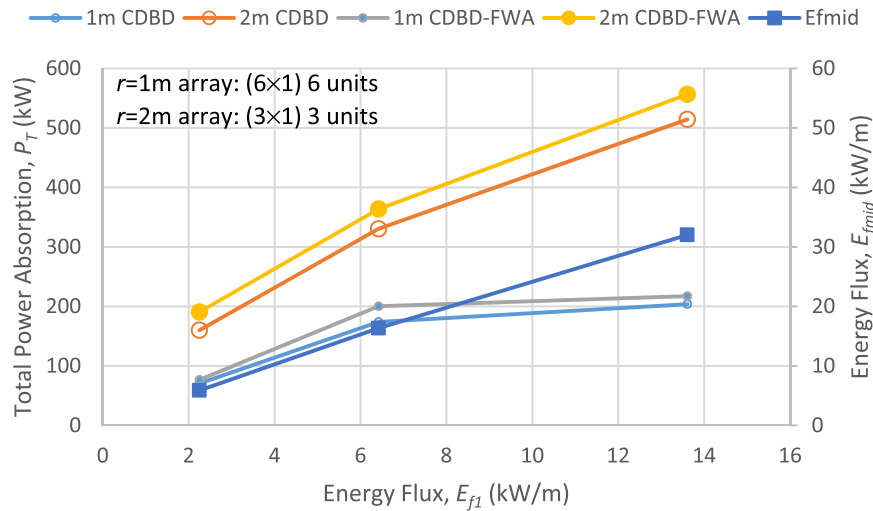


Fig. 13. Comparison of in-line WEC arrays.

place another in-line array somewhere towards the shore from the first array to harness the remaining energy. In this way, multiple in-line arrays could be placed in the down wave direction to form a rectangular array. Illustrated in Fig. 14 are the structures of rectangular arrays, which could be formed inside the concentrator by using the in-line arrays. There are three in-line arrays in both rectangular arrays formed using $r = 1$ m and 2 m elements. The spacing between the in-line arrays are 12 m and only the latter part of the uniform channel, where the waves are steadier, is occupied.

The spectrum of the down wave is likely to be different from that of the incident wave due to the radiations from the buoys and reflections from not only the buoys but also from the walls of the concentrator. Therefore, in order to estimate the total power absorption of the whole rectangular array, such effects are again assumed to be negligible. The validity of this assumption is dependent on the peak frequency of the spectrum and the spacing among elements. Note again that it has been mentioned in (McIver and Bennett, 1993; Eriksson, 2005; Gräter, 2016) that the wave diffractions are usually negligible for low frequency waves such as sea waves and when the buoy dimensions are small with respect to the wavelength. Further, the reflections would again be negligible if inter-elemental spacings are high enough. Therefore, the estimation of power absorption by neglecting the diffraction would not be very crude because it is rather possible to satisfy the associated conditions in practice. Note however that the energy flux of the incident wave is decreased by not only the energy absorbed by the WEC array but also the radiation losses. The average radiation loss could be found for each wave component with frequency ω as (Eriksson, 2005; Gräter, 2016),

$$P_{rd}(\omega) = 0.5R(\omega)|\dot{z}(\omega)|^2 \quad (40)$$

In this paper, the remaining energy flux of the down wave is found by subtracting the total power absorbed by the corresponding in-line array and the total radiation loss from the energy flux of the incident wave for

each wave component. This step-by-step reduction in energy flux after each in-line array is also evident from the experimental results presented in (Stratigaki et al., 2014). It can be noted that the energy flux has a tendency to be constant between a pair of in-line arrays with some exceptions. Therefore, it would be possible to evaluate the power absorption of each element in the next in-line array by solving the motion equation in (8) again. However, in this paper, a simplified approach is used for this purpose. In this case, the percentage power absorption in each wave frequency component in the second and third arrays are considered to be the same as those in the first in-line array. If the remaining energy flux at those second or third in-line arrays is too small to provide even the radiation losses, the power absorption is considered to be zero. The overall power absorption of the rectangular arrays estimated in this way are illustrated in Fig. 15 for both WEC types. At each sea state, the overall power absorption possible with rectangular arrays formed from CDBD WEC units are somewhat less than those formed from CDBD-FWA WEC units.

The increase in power absorption possible with a rectangular array is not a factor of the number of constituent in-line arrays. This is because the available energy flux becomes smaller and smaller in the down wave after each in-line array. Listed in Table 20 are the power absorptions possible with one in-line array (3 units) and a rectangular array (9 units) formed from both type of WECs with $r = 2$ m. Note that the scale of increase is smaller in the low sea state while it is higher in the high sea state. For example, with CDBD-FWA units, 1 MW of power absorption is possible with the 9-element rectangular array but only 0.56 MW is possible with a single 3-element in-line array, when the total power captured into the concentrator is 1.36 MW in Sea state 3. However, this difference in power absorption is smaller in Sea state 1.

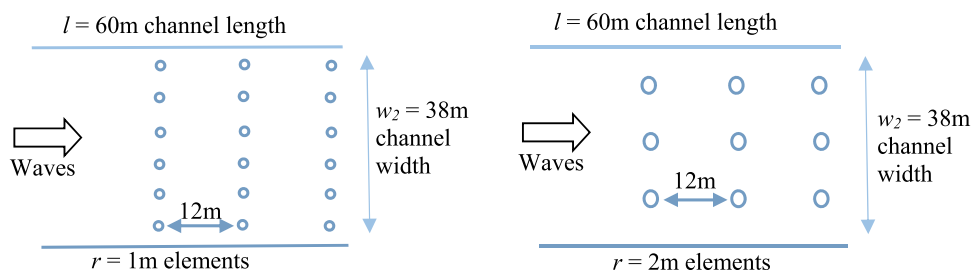


Fig. 14. Structure of the rectangular arrays inside the concentrator.

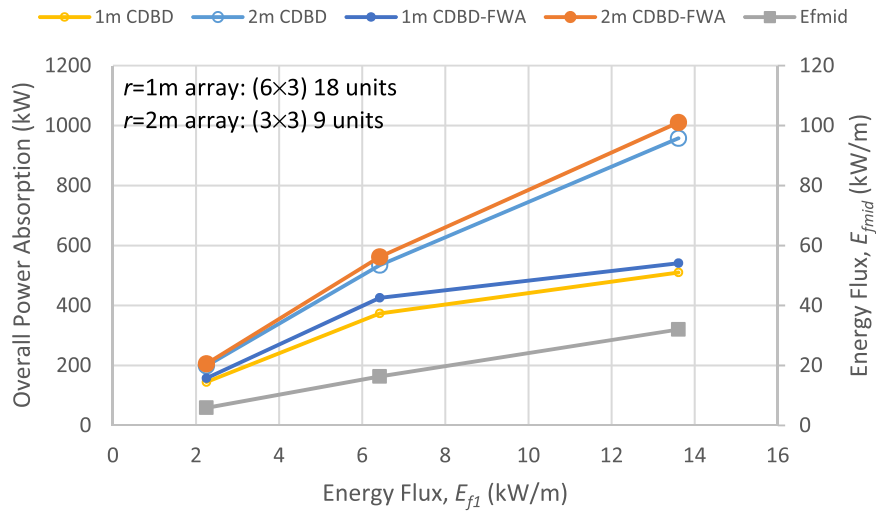


Fig. 15. Comparison of rectangular arrays.

Table 20

Comparison of in-line arrays and rectangular arrays with $r = 2$ m elements.

Nearshore Sea state	Nearshore energy flux, E_{f1} (kW/m)	Mid-channel energy flux, E_{fmid} (kW/m)	Total available power (kW)	Overall power absorption of CDBD- WEC arrays (kW)		Overall power absorption of CDBD- FWA WEC arrays (kW)	
				In-line (3 elements)	Rectangular (9 elements)	In-line (3 elements)	Rectangular (9 elements)
1	2.25	5.88	225	159.9	198.6	190.6	205.7
2	6.42	16.34	642	330.4	534.4	363.7	562.1
3	13.6	32	1360	514.3	957.6	556.4	1010.2

7. Discussion

A point absorber type WEC, in which the mechanical components and the electrical generator are mounted above the sea level, has been described and analyzed in this paper. The motion equation of the top-mounted WEC has been solved using a rather new approach. First, F_w has been represented using the wave dynamic pressure so that an approximate 2nd order transfer function of the WEC system could be formed. Hence, it has been possible to check the stability and the frequency response of the system. Then, the time average and/or the spectral average of the power absorption has been evaluated using the frequency response for a given spectrum of wave components. It has been shown that some coefficients of the frequency response are frequency dependent requiring a trial-and-error procedure to find the natural frequency. It has been further shown that down-shifting the natural frequency using m_1 alone to match the wave spectrum is not satisfactory for irregular waves.

The effect of the phase distribution of the wave spectrum has been considered in this paper distinguishing two cases. The wave spectrum may be a representation of a natural composite wave which is a summation of real wave components propagating with different heights, wavelengths and phases, or otherwise a Fourier decomposition of a single non sinusoidal waveform such as a higher order Stokes wave. In the latter case, the wave spectrum is purely a mathematical expansion with a stationary phase distribution. Hence, the power absorption would be steadier and the spectral estimate would be reliable. In the former case, real time variations of not only the phase distribution but also the height distribution could be expected. This phenomenon could also be seen as a real time variation of the sea state for example due to local winds. In this paper, the phase distribution has been considered to be a variable in order to account for natural composite waves. Therefore, in

the worst case, depending on the prevailing phase distribution, the fluctuation of the time average of power absorption ($P(\epsilon_N)$) might be high and the buoy might become airborne from time to time. Therefore, in this paper, the system response has been reshaped optimally by adjusting both m_1 and B_{PTO} (in CDBD-FWA WEC) such that the spectral estimate of the power absorption (P_{ave}) and the possibility of the buoy becoming airborne are compromised. In addition, the possibility of the buoy becoming submerged in water has been considered too. In this case, the power absorptions are not affected but ER is decreased by a small amount in all the sea states.

In practice, m_1 and B_{PTO} needs to be set to the optimum values whenever there is a change in the sea state. If a given sea state is in effect for a fairly long time such as for a few hours, m_1 could be adjusted manually. If the flywheel is made up of several disks, m_1 could be increased or decreased step by step, by engaging and disengaging the disks as appropriate. The value of B_{PTO} could be varied by changing the impedance of the load connected to the generator. If the load is a battery bank, the range of the internal resistances must be such that the expected range of B_{PTO} could be realized. Then, one or more units of the battery bank could be turned on or off to set B_{PTO} suitable for the prevailing sea state. Both these adjustments could be made trial and error simply by checking the reading in a power meter connected to the electrical system. Otherwise, the real time sea state information could be measured by placing another buoy equipped with measuring instruments and a data acquisition and transmission system. Based on such real time sea state measurements, m_1 and B_{PTO} could be adjusted faster and more accurately. However, if the sea state changes are more frequent, this open loop adaptive system could not be implemented manually. In this case, by providing the real time sea state information, m_1 could be changed using a device which is similar to an automatic gearbox while the internal resistance of the battery bank and B_{PTO} could

be changed using an electronic circuit.

With this open loop adaptive control system, WEC system parameters (m_1 and B_{PTO}) are set to suit the prevailing sea state by using a table look-up procedure. The optimal values of m_1 and B_{PTO} for each possible sea state are listed in this table. As long as the sea state is stationary for a period which is reasonably higher than the time taken for adjustments, this method would be satisfactory. However, the time taken for these adjustments should not be higher than a few minutes even if it is carried out manually. If in a certain site, sea state changes are more rapid in general, that site is not suitable for a WEC. Note that this is not a closed loop control system in which the absorbed power is used to vary the system parameters. In such a system, instabilities may be observed requiring further analysis and experimentations. However, in the open loop control system described in this paper, the sea state information which is used to choose the optimum set of m_1 and B_{PTO} , is obtained from an external system. This external system is not physically connected to the WEC system and has no direct influence on the WEC.

In the results section of this paper, the spectral estimates of power absorption and the corresponding reliability of the systems with response reshaping (CDBD-FWA) and with no response reshaping (CDBD) have been presented for deep sea waves (JONSWAP model), nearshore waves (TMA) and concentrated nearshore waves. In some cases, short-term time averages and phase averages of power absorptions have been given for comparison. In all cases, the power absorption of CDBD-FWA has been higher but the estimation reliability has been somewhat lower than that of CDBD. In highlighting, the power absorption of CDBD-FWA WEC with a buoy radius of 2 m is about 8% - 18% higher than that of CDBD WEC in the three wave environments considered. It is about three times higher with both WECs in the concentrated TMA waves than in normal TMA waves. Finally, WEC arrays have been considered in order to maintain the power absorption at a certain useful level. The total power absorption of an in-line array and a rectangular array suitable for the wave concentrator has been estimated somewhat crudely. In highlighting again, the overall power absorption of a 9-element rectangular array is found to be only 1.8 (instead of 3) times higher than that of the 3-element in-line array in the high sea state of concentrated waves. In those sea states with lower energy, this factor is even smaller.

The power absorptions of the top-mounted WECs with and without response reshaping described in this paper have been compared to those in other publications wherever appropriate. For example, the point absorber type WEC system, which is supported by a spring fixed to an offshore oil and gas platform, described in (Pastor and Liu, 2014) is somewhat similar to a top-mounted system described in this paper. The main difference is that the shape of the buoy in the former is not cylindrical but conical. The comparisons in similar JONSWAP sea states have indicated that the power absorption of the system described in this paper is higher in the high sea state with high energy flux but lower in the low sea state. This increase in power absorption in the low sea state could be attributed to the conical shape of the buoy considered in (Pastor and Liu, 2014), with which the dynamic range and the velocity of the buoy has been relatively enhanced. The power absorption of a point absorber with a hydraulic PTO system described in (Sinha et al., 2016) and CDBD-FWA system described in this paper are more or less the same in a similar TMA sea state.

Although a number of other publications on point absorber type WECs have been cited in this paper, they were found to be different with respect to the parameters such as M , r , d , K_s and B_{PTO} . In addition, those WECs are of bottom-mounted type and the results are based on different standard sea states or real sea trials. Hence, any precise quantitative comparisons have not been possible. However, in a bottom-mounted system (Sjökvisst et al., 2014), the power absorption is 19 kW when $r = 2$ m and $B_{PTO} = 124$ kNs/m for a monochromatic wave with $\omega =$

1 rad/s. With the same three parameters, the power absorption of the top-mounted system is found to be between 8.1 kW (lower bound) and 32.5 kW (upper bound). However, there are significant differences in M and d . According to this crude comparison for a monochromatic wave, it can be concluded that at least the order of the power estimates is similar in the top-mounted and bottom mounted systems. A similar bottom-mounted WEC (Potapenko et al., 2021) has been shown to absorb 16 kW by using real sea data with a buoy of $r = 2$ m and $M = 3000$ Kg. It has been shown in this paper that about seven times more power could be absorbed by using a buoy with seven times more mass in a JONSWAP sea state with similar H_{m0} and T_e .

The power estimates taken in the real sea by using measured sea state data or by installing prototypes (Eriksson et al., 2007; Engström et al., 2009; Silvia et al., 2013; Lejerskog et al., 2015) are found to be comparatively smaller. One reason for this is that the wave heights in the test sites are relatively low and the power absorption is decreased with the square of the wave heights. The other reason is that the phase distribution of the wave components in the real sea may not have been favorable or stationary. Therefore, the short-term power absorption may have been relatively low due to constructive additions of wave heights and breaking or otherwise due to cancellations of wave heights. In this way, even the long term power absorption could be small. In practical terms, it is possible that once a WEC system is constructed and tested for the first time, the absorbed power over some time could be much smaller than the expected spectral average (P_{ave}). Therefore, it is not sensible to abandon the prototype based on such short-term power production because the device may produce expected levels of power even in the same sea state but with a different phase distribution which occurs at other times, a few days or weeks later. Even if the sea state changes to a favorable one after a long time period, the short-term power production would still fluctuate between the corresponding minimum and the maximum as illustrated for JONSWAP waves in Section 6.1. It is true that seasonal variations of the wave parameters such as ω_p , H_{m0} , E_{f0} and E_{f1} can be predicted well in advance using the databases containing sea state information over decades (Barstow et al., 2009; Mørk et al., 2010) and that any short-term changes could be predicted a few days ahead (Coe et al., 2021). However, not only for different point absorber type WEC structures but also for some other types of WECs, there may be favorable phase distributions which result in absorption of high power. This fact in turn makes comparison of different types of WECs a tedious and a time consuming task. It also implies that it may take a long time in practice to measure the power absorption of a prototype and to approve or abandon it because it may take a long time for the favorable, unfavorable or expected phase distributions to occur.

It has been explained in this paper that the estimation reliability is a measure of not only the number of time intervals during which the buoy is likely to become airborne or submerged but also the possibility of wave breaking due to the constructive addition of wave heights. If the prevailing phase distribution is unfavorable, the short-term power production would be small due to wave breaking or low levels of wave height. However, in practice, such problems are normally avoided by using proper site selection methods based on long term observations and measurements. In a certain sea site, if wave breaking is more common or the aggregate wave height is small, that site should not be chosen for wave energy harnessing.

The cost of manufacturing, installation, testing and maintenance of a top-mounted point absorber type WEC placed in the nearshore would be relatively small. This could be qualitatively justified easily by considering the reasons explained in Sections 1 and 2. An absolute cost could only be found by considering such parameters as the availability of materials, equipment and human resources locally, markets in other countries, local manufacturing capabilities, trade agreements, taxes and subsidies. Additionally, a number of other parameters which should be

considered in a cost evaluation, has been described in (Sandberg et al., 2016). Therefore, any quantitative analysis of the absolute cost is not included in this paper.

8. Conclusions

The advantages of a top-mounted point absorber type WEC placed in the nearshore or inside a wave concentrator in the nearshore are numerous. Some of the advantages can be validated theoretically, semi-theoretically or semi-empirically. Some others can be validated only logically until a real prototype is installed and tested. The power absorption of the top-mounted point absorber type WEC appears to be somewhat higher than that of most other point absorber type systems in monochromatic waves. With a 2 m buoy, the average power absorption with frequency response reshaping in 30 kW/m JONSWAP waves is about 114 kW, which is about 29 kW higher than that of a somewhat similar system. With a 1 m buoy, it is more or less the same (28 kW) as that of a point absorber system with a hydraulic PTO in 13.6 kW/m TMA waves. With a 2 m buoy, the power absorption possible by concentrating the same TMA waves is about 186 kW. In all the TMA sea states considered, the increase in power absorption possible by wave concentration is almost three times, which is close to the concentration gain of 2.6. A high proportion of the remaining energy flux of the concentrated waves can be harnessed by placing a linear longshore array or a rectangular array with enough spacing among WEC units. In any such deployment, the power absorption possible with response reshaping is considerably higher than that without response reshaping. In this way, the total power absorption would be about 556 kW with a 3-element longshore in-line array and about 1 MW with a 9-element rectangular array in the same concentrated TMA waves. The short-term power absorption is dependent on the prevailing phase distribution of the sea wave spectrum. At times, this can be very small due to the intervals of either airborne incidences, submersions, wave breaking incidences with constructive addition of wave components or diminishing waves with cancellation of wave components, even in the same sea state with the same energy flux. Hence, WECs should only be installed at proper sites and should not be underestimated based on short-term tendencies.

There are mainly three limits of this approach of wave energy conversion. Firstly, the size and the mass of the buoy cannot be increased at will. The power absorption would be decreased and the wave scattering and diffraction would be increased with large buoys. Secondly, this approach would be less effective in long waves with high wavelengths and periods. Thirdly, it is very sensitive to the prevailing phase distribution. If the phase changes are too rapid and unfavorable phase distributions are more likely, the power absorption could be small. In continuing this research, it will be worthwhile to consider the ways of increasing power absorption of WEC arrays by making use of the scattering and diffraction constructively. It will also be useful to estimate the power absorption in those time periods during which the buoy is in-air or submerged and to include the vertical motion of the chain caused by the surge and sway too in the analysis. The cost of manufacturing, installation, testing and maintenance of a top-mounted WEC placed in the nearshore will be relatively small due to a number of reasons. However, it will also be useful to estimate the absolute cost.

CRedit authorship contribution statement

K.D.R. Jagath Kumara: Conceptualization, Methodology, Validation, Formal analysis, Visualization, Writing – original draft, Writing – review and editing.

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References

- Adachi, T., Takahara, K., 2016. Analysis and measurement of damping characteristics of linear generators. *Int. J. Appl. Electromagn. Mech.* 52, 1503–1510. <https://doi.org/10.3233/JAE-162166>.
- Aderinto, T., Li, H., 2019. Review on power performance and efficiency of wave energy converters. *In: Energies*, 12(22). MDPI. doi:10.3390/en12224329.
- Akylas, T.R., Mei, C.C., 2001–2014. MIT Open Courseware - Modules on waves in Fluids. Massachusetts Institute of Technology, U.S.A.
- Allsop, W., Bruce, T., Alderson, J., Ferrante, V., Russo, V., Vicinanza, D., Kudella, M., 2014. Large scale tests on a generalized oscillating water column wave energy converter. *Proc. HYDRALAB IV Jt. Use Meet., Lisbon*. In: <https://hydralab.eu/uploads/proceedings/TA-FZK-02.pdf>.
- Barstow, S., Mørk, G., Lønseth, L., & Mathisen, J.P., (2009) “WorldWaves wave energy resource assessments from the deep ocean to the coast,” in *Proc. 8th European Wave and Tidal Energy Conference*, Uppsala, Sweden, p. 149–159.
- Basco, D.R., & Yamashita, T., (1987) “Toward a simple model of the wave breaking transition region in surf zones”, *20th International Conference on Coastal Engineering*, 9–14 Nov., Taipei, Taiwan (1987 American Society of Civil Engineers).
- Bendykowska, G., Werner, G., 1988. Transformation of shallow water wave spectra. *Coast. Eng.* 1 (21), 612–623. <https://doi.org/10.9753/icce.v21.44>.
- Bergdhal, L., (2009), “Comparison of measured shallow-water wave spectra with theoretical spectra”, *Proceedings of the 8th European Wave and Tidal Energy Conference*, Uppsala, Sweden. p. 100–105.
- Brask, A.K., (2015), Control and Estimation of Wave Energy Convertors, Master Thesis, Norwegian University of Science and Technology (NTNU), Trondheim, Norway.
- Camilla Thomson, R., Chick, J.P., Harrison, G.P., 2018. An LCA of the Pelamis wave energy converter. *Int. J. Life Cycle Assess.* 24 (1), 51–63. <https://doi.org/10.1007/s11367-018-1504-2>.
- Chang, P., Melville, W.K., Miles, J.W., 1979. On the evolution of a solitary wave in a gradually varying channel. *J. Fluid Mech.* Vol. 95 (3), 401–414. <https://doi.org/10.1017/S002211207900152X>.
- Coe, R.G., Bacelli, G., Forbush, D., 2021. A practical approach to wave energy modelling and control. *Renewable and Sustainable Energy Reviews*. Elsevier, pp. 1–11. <https://doi.org/10.1016/j.rser.2021.110791>.
- Dupont, G., Kimmoun, O., Molin, B., (2012), “Experimental and numerical analysis of the wave propagation through a narrow channel in a wave flume”, In *Proc. 27th International workshop on water waves and floating bodies*, Copenhagen, Denmark, April, p. 49–52.
- Elgar, S., Guza, R.T., Seymour, R.J., 1985. Wave group statistics from numerical simulations of a random sea. *Appl. Ocean Res.* 7 (2), 93–96. [https://doi.org/10.1016/0141-1187\(85\)90039-2](https://doi.org/10.1016/0141-1187(85)90039-2).
- Engström, J., 2011. Hydrodynamic Modelling for a Point Absorbing Wave Energy Converter. PhD Thesis, Uppsala University, Sweden.
- Engström, J., Eriksson, M., Goteman, M., Isberg, J., Leijon, M., 2013. Performance of large arrays of point absorbing direct-driven wave energy converters. *J. Appl. Phys.* 114. <https://doi.org/10.1063/1.4833241>.
- Engström, J., Eriksson, M., Isberg, J., Leijon, M., 2009. Wave energy converter with enhanced amplitude response at frequencies coinciding with Swedish west coast sea states by use of a supplementary submerged body. *J. Appl. Phys.* 106, 064512. <https://doi.org/10.1063/1.3233656>.
- Engström, J., Kurupath, V., Isberg, J., Leijon, M., 2011. A resonant two body system for a point absorbing wave energy converter with direct driven linear generator. *J. Appl. Phys.* 110, 124904. <https://doi.org/10.1063/1.3664855>.
- Eriksson, M., Isberg, J., Leijon, M., 2005. Hydrodynamic modelling of a direct drive wave energy converter. In: *International Journal of Engineering Science*. Elsevier, pp. 1377–1387. <https://doi.org/10.1016/j.ijengsci.2005.05.014>.
- Eriksson, M., Waters, R., Svensson, O., Isberg, J., Leijon, M., 2007. Wave power absorption: experiments in open sea and simulation. *J. Appl. Phys.* 102, 084910. <https://doi.org/10.1063/1.2801002>.
- Faizal, M., Ahmed, M.R., Lee, Y.H., 2014. A design outline for floating point absorber wave energy converters. *Advances in Mechanical Engineering*, 6. Hindawi, 846097. <https://doi.org/10.1155/2014/846097>.
- Felix, A., Hernández-Fontes, J.V., Lithgow, D., Mendoza, D., Posada, G., Ring, M.R., Silva, R., 2019. Wave energy in tropical regions: deployment challenges, environmental and social perspectives. *J. Mar. Sci. Eng.* MDPI 7, 219–239. <https://doi.org/10.3390/jmse7070219>.
- Folley, M., Lamont-Kane, P., Frost, C., 2023. Linear hydrodynamic model of rotating lift-based wave energy converters. *Int. Mar. Energy J.* 6 (2), 37–44. <https://doi.org/10.36688/imej.6.37-4>.
- Goda, Y., 2010. Reanalysis of regular and random breaking wave statistics. *Coast. Eng. J.* 52 (1), 71–106. <https://doi.org/10.1142/S0578563410002129>.

- Grüter, L., (2016) "Experimental and Numerical Analysis of a Wave Energy Conversion Buoy", Diplomarbeit, Technische Universität Berlin und Universidad Austral de Chile.
- Halder, P., Hyung Rhee, S., Samad, A., 2017. Numerical optimization of wells turbine for wave energy extraction. *Int. J. Nav. Archit. Ocean Eng.* 9, 1–24. <https://doi.org/10.1016/j.ijnaoe.2016.06.008>.
- Hansen, J.B., 1990. Periodic waves in the surf zone: Analysis of experimental data. *Coast. Eng.* Vol. 14 (1), 14–41. [https://doi.org/10.1016/0378-3839\(90\)90008-K](https://doi.org/10.1016/0378-3839(90)90008-K).
- Hasselmann, K., et al., (1973), "Measurements of wind wave growth and swell decay during the joint sea wave project (JONSWAP)," Report No. 12, Deutshes Hydrographisches Institut, Hamburg, Germany.
- Hong, Y., Waters, R., Boström, C., Eriksson, M., Engström, J., Leijon, M., 2014. Review on electrical control strategies for wave energy converting systems. *Renewable and Sustainable Energy Reviews*, 31. Elsevier, pp. 329–342. <https://doi.org/10.1016/j.rser.2013.11.053>.
- Hughes, S.A., (1984), "The TMA shallow-water spectrum description and applications", Technical report, CERC-84-7, Coastal Engineering Research Centre, US Army Engineer Waterways Experiment Station, Vicksburg, Mississippi, U.S.A.
- Ibrahim, W.I., Mohamed, M.R., Ismail, R.M.T.R., Leung, P.K., Xing, W.W., Shah, A.A., 2021. Hydrokinetic energy harnessing technologies: A review. In: *Energy Reports*. Elsevier, pp. 2021–2042. <https://doi.org/10.1016/j.egy.2021.04.003>.
- Inch, K.W., 2014. Surf zone hydrodynamics. *Br. Soc. Geomorphol. Geomorphol. Tech. Sec. 2.3*.
- Jagath Kumara, K.D.R., Dias, D.D., 2015. Near shore wave manipulation for electricity generation. *WASET Int. J. Energy Power Eng.* Vol. 9 (7), 683–692.
- Jagath Kumara, K.D.R., Dias, D., Nawagamuwa, R.L., Weerasinghe, W.M.A.R., 2018. Wave energy enhancement for nearshore electricity generation. *Eng. J. Inst. Eng.* Vol. LI (2), 43–52.
- Jagath Kumara, K.D.R., 2022. On the placement of a wave manipulator suitable for energy harnessing in the nearshore. *Cogent Eng., Taylor Fr.* Vol. 9 (1) (September). doi:10.1080/23311916.2022.2124636.
- Kamen, E.W., Heck, B.S., 2000. *Fundamental of Signals and Systems Using the WEB and MATLAB*. Prentice-Hall.
- Kaselim, M., & Delikaraoglou, D., (2018), "Estimating the prospects of wave energy potential in Eastern Mediterranean using multi-mission satellite altimeter data", In "Quod Erat Demonstrandum – In quest of the ultimate geodetic insight", Honorary volume dedicated to Professor Emeritus A. Dermanis, School of Rural and Surveying Engineering, University of Thessaloniki, Greece Eds. A. Fotiou & D. Rossikopoulos, Ziti Publisher, Thessaloniki, Greece, p. 221–238.
- Kimmoun, O., Branger, H., 2007. A particle image velocimetry investigation on laboratory surf-zone breaking waves over a sloping beach. *J. Fluid Mech.* 588, 353–397. <https://doi.org/10.1017/S0022112007007641>.
- Kitaigorodskii, S.A., Krasitskii, V.P., Zaslavskii, M.M., 1975. On Phillip's theory of equilibrium range in the spectra of wind-generated gravity waves. *J. Phys. Oceanogr.* 5, 410–420. [https://doi.org/10.1175/1520-0485\(1975\)0052.0.CO;2](https://doi.org/10.1175/1520-0485(1975)0052.0.CO;2).
- Lejerskog, E., Bostrom, C., Hai, L., Waters, R., Leijon, M., 2015. Experimental results on power absorption from a wave energy converter at the Lysekil wave energy research site. *Renewable Energy*. Elsevier, pp. 9–14. <https://doi.org/10.1016/J.RENENE.2014.11.050>.
- Maliyadda, S.D.K., Wijeratne, W.M.C.R., Zoysa, S.R.L.M., Dias, D., & Jagath Kumara, K. D.R., (2014), "Manipulation of near-shore sea waves for electricity generation: modelling a wave concentrator," in *Proc. 5th International Conference on Sustainable Built Environment*, ICSBE 2014, Kandy, Sri Lanka, Vol. 3, (<https://www.icsbe.org>), 206–216, 12–14 Dec.
- Malta, E.B., Gonçalves, R.T., Matsumoto, F.T., Pereira, F.R., Fajarra, A.L.C., Nishimoto, K., (2010), "Damping coefficient analyses for floating offshore structures", *Proceedings of the ASME 29th International Conference on Ocean, Offshore and Arctic Engineering OMAE2010-20093*, Shanghai, China, 9, p. 83–89. doi:10.1115/OMAE2010-20093.
- McIver, P., Bennett, G.S., 1993. Scattering of water waves by axisymmetric bodies in a channel. *J. Eng. Math.* Vol. 27, 1–29. <https://doi.org/10.1007/BF00128970>.
- Mehlum, E., 1986. TAPCHAN. In: Evans, D.V., de Falcao, A.F.O. (Eds.), *Hydrodynamics of Ocean Wave Energy Utilization*. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-82666-5_3.
- Mørk, G., Barstow, S., Kabuth, A., & Teresa Pontes, M., (2010), "Assessing the global wave energy potential", In *Proc. 29th International Conference on Ocean, Offshore Mechanics and Arctic Engineering*, Shanghai, China, June 6 – 11, OMAE2010 - 20473. doi:10.1115/OMAE2010-20473.
- Moulton, M., Elgar, S., Raubenheimer, B., Warner, J.C., Kumar, Nirmimesh, 2017. Rip currents and alongshore flows in single channels dredged in the surf zone. *J. Geophys. Res.: Oceans* Vol. 122 (5). <https://doi.org/10.1002/2016JC012222>.
- Murai, M., Li, Q., Funada, J., 2021. Study on power generation of single point absorber wave energy converters (PA-WECs) and arrays of PA-WECs. In: *Renewable Energy*. Elsevier, pp. 1121–1132. <https://doi.org/10.1016/j.renene.2020.08.124>.
- Mutombo, N.M.A., Numbi, B.P., 2022. Absorbed power density approach for optimal design of heaving point absorber wave energy converter: a case study of Durban sea characteristics. *J. Energy South Afr.* Vol. 33 (No. 1), 52–67. <https://doi.org/10.17159/2413-3051/2022/v33i1a8362>.
- Parmeggiani, S., Peter Kofoed, J., Friis-Madsen, E., 2013. Experimental study related to the mooring design for the 1.5 MW Wave Dragon WEC demonstrator at DanWEC. *Energ., MDPI* 6 (4), 1863–1886. <https://doi.org/10.3390/en6041863>.
- Pastor, J., Liu, Y., 2014. Frequency and time domain modelling and power output for a heaving point absorber wave energy converter. In: *International Journal of Energy and Environmental Engineering*. Springer, p. 101. <https://doi.org/10.1007/s40095-014-0101-9>.
- Pecher, A., & Peter Kofoed, J. (Editors), (2017), *Handbook of Ocean Wave Energy*, Ocean Engineering and Oceanography, Vol. 7, Springer Open, ISSN 2194-640X (electronic).
- Peter Kofoed, J., Frigaard, P., Friss-Madsen, E., Sørensen, H.C., 2006. Prototype testing of the wave energy converter wave dragon. In: *Renewable Energy*. Elsevier, pp. 181–189. <https://doi.org/10.1016/j.renene.2005.09.005>.
- Phillips, O.M., 1958. The equilibrium range in the spectrum of wind-generated ocean waves. *J. Fluid Mech.* 4, 426–434. <https://doi.org/10.1017/S0022112058000550>.
- Pierson, W.J., Moskowitz, L., 1964. A proposed spectral form for fully developed wind seas based on the similarity theory of J. Geophys. Res. 69, 5181–5190. <https://doi.org/10.1029/JZ069I024P05181>.
- Potapenko, T., Burchell, J., Eriksson, S., Temiz, I., 2021. Wave energy converter's slack and stiff connection: Study of absorbed power in irregular waves. In: *Energies*. MDPI, p. 7892. <https://doi.org/10.3390/en14237892>.
- Power, H.E., Hughes, M.G., Aagaard, T., Baldock, T.E., 2010. Nearshore wave height variation in unsaturated surf. *J. Geophys. Res.* 115. <https://doi.org/10.1029/2009JC005758>.
- Ringwood, J., 2008. Practical challenges in harvesting wave energy. In: *ECOR Symposium*, S. St. John's, Newfoundland, Canada.
- Robert, A.D., 2024. *Water Wave Propagation in Jettied Channels*. Centre for Applied Research, Department of Civil Engineering, University of Delaware, Newark, DE 19716, U.S.A.
- Rueda-Bayona, J.G., Guzmán, A., Eras, J.J.C., 2020. Selection of JONSWAP spectra parameters during water-depth and sea-state transitions. *J. Waterw. Port. Coast. Ocean Eng.* ASCE 146 (6), 1–13. [https://doi.org/10.1061/\(ASCE\)WW.1943-5460.0000601](https://doi.org/10.1061/(ASCE)WW.1943-5460.0000601), 04020038.
- Rusch, C.J., Green, D.R., Sager, D.S., McGarry, J.D., Morasch, J.T., Downs, J.A., 2015. "Dynamics of a point absorber wave energy converter", ME495.2015 Spring Report, UW WEC Team, Department of Mechanical Engineering, University of Washington, Seattle, WA 98195–2600 U.S.A.
- Sandberg, A.B., Klemetsen, E., Muller, G., Andres, A.D., Maillet, J., 2016. Critical factors influencing viability of wave energy converters in off-grid luxury resorts and small utilities. *Sustain., MDPI* 8, 1274. <https://doi.org/10.3390/SU8121274>.
- Silvia, B., Miquel, A.M., Antonini, A., Passoni, G., Archetti, R., 2013. Modelling of a point absorber for energy conversion in Italian seas. *Energies* 6, 3033–3051. <https://doi.org/10.3390/EN6063033>.
- Sinha, A., Karmakar, D., Guedes Soares, C., 2016. Shallow water effects on wave energy converter with hydraulic power take-off system. In: *The International Journal of Ocean and Climate Systems*. SAGE, pp. 108–117. <https://doi.org/10.1177/1759313116649966>.
- Sjökvist, L., Krishna, R., Rahm, M., Castellucci, V., Hagnestål, A., Leijon, M., 2014. On the optimization of point absorber buoys. *J. Mar. Sci. Eng.* 2, 477–492. <https://doi.org/10.3390/JMSE2020477>.
- Sørensen, O.R., Schaffer, H.A., Madsen, P.A., Deigaard, R., 1994. Wave breaking and induced nearshore circulations. *Coast. Eng.* 1, 2583–2594. <https://doi.org/10.1061/9780784400890.188>.
- Stratigaki, V., Troch, P., Forehand, D., Stallard, T., Peter Kofoed, J., Folley, M., Benoit, M., Babarit, A., Kirkegaard, J., Sánchez, D.G., Bosscher, L.D., Rauwoens, P., Elsässer, B., Lamont-Kane, P., McCallum, P., McNatt, C., Angelelli, E., Pecher, A., Moreno, E. C., Bellew, S., Dombre, E., Charayre, F., Carstensen, S., Stansby, P., Vantorre, M., (2014), "Physical modelling of wave energy converter arrays in a large-scale wave basin: The WECwakes project" *Proceedings of the HYDRALAB IV Joint User Meeting*, Lisbon, Portugal.
- Svendsen, I.A., 1983. Wave heights and set-up in a surf zone. *Coast. Eng.* Vol. 8 (4), 303–330. [https://doi.org/10.1016/0378-3839\(84\)90028-0](https://doi.org/10.1016/0378-3839(84)90028-0).
- Svendsen, Ib.A., 2006. *Introduction to nearshore hydrodynamics*. World Scientific Publishing Co. Pte. Ltd.
- Svendsen, Ib.A., 1992. "Hydrodynamics of the Surf Zone", Centre for Applied Coastal Research, University of Delaware, Newark DE 19716, U. S. A. (2024).
- Szmidt, K., Hedzielski, B., 2016. Transformation of long waves in a canal of variable section. In: *Archives of Hydro-Engineering and Environmental Mechanics*. De Gruyter Open, pp. 3–18. <https://doi.org/10.1515/heem-2016-0001>.
- Tan, M., Cen, Y., Yang, Y., Liu, X., Si, Y., Qian, P., Zhang, D., 2021. Power absorption modelling and analysis of a multi-axis wave energy converter. In: *IET renewable power generation*. Wiley, pp. 3368–3384. <https://doi.org/10.1049/rpg.2.12277>.
- Thomas, S., Giassi, M., Göteman, M., Hann, M., Ransley, E., Isberg, J., Engström, J., 2018. Performance of a direct-driven wave energy point absorber with high inertia rotational power take-off. In: *Energies*. MDPI, p. 2332. <https://doi.org/10.3390/en1092332>.
- Thurman, H.V., & Trujillo, A.P., 2001. *Essentials of Oceanography*, 7th edition, Ch. 8: Waves and water dynamics, Prentice-Hall. <http://www.prenhall.com/thurman>.
- Tissier, M., Bonneton, P., & Ruessink, G., (2017), "Infragravity waves and bore merging", In Aagaard, T., Deigaard, R., & Fuhrman, D. (Eds.), *Proceedings of Coastal Dynamics 2017*, Helsingør, Denmark, p. 451–460. [Paper No. 209]. <http://coastaldynamics2017.dk>.
- Van Dongeren, A., Battjes, J., Janssen, T., Noorloos, J.V., Steenhauer, K., Steenbergen, G., Reniers, A., 2007. Shoaling and shoreline dissipation of low-frequency waves. *J. Geophys. Res.* Vol. 112, C02011. <https://doi.org/10.1029/2006JC003701>.

- Wang, L., Lin, M., Tedeschi, E., Engström, J., Isberg, J., (2020), "Improving electric power generation of a standalone wave energy converter via optimal electric load control", arXiv:2009.12485v1, Cornell University, U.S.A. doi:10.1016/J.ENERGY.2020.118945.
- Watabe, T., 2008. Utilization of the ocean wave energy. Fuji Print Press Co. Ltd, Japan.
- Waters, R., 2008. Energy from ocean waves. PhD Thesis, Uppsala University, Sweden.
- Wendt, F., Yi-Hsiang Y., Nielsen, K., Ruehl, K., Bunnik, T., Touzon, I., Nam, B.W., Jeong Seok K., Kyong-Hwan K., Janson, C.E., Jakobsen, K.R., Crowley, S., Vega, L., Rajagopalan, K., Mathai, T., Greaves, D., Ransley, E., Lamont-Kane, P., Sheng, W., Costello, R., Kennedy, B., Thomas, S., Heras, P., Bingham, H., Kurniawan, A., Kramer, M.M., Ogden, D., Girardin, S., Babarit, A., Wuillaume, P.Y., Steinkey, D., Roy, A., Beatty, S., Schofield, P., Jansson, J., Hoffman, J., (2017), "International energy agency ocean energy systems task 10 (IEA OES Task 10) wave energy converter modelling verification and validation", Proceedings of the 12th European Wave and Tidal Energy Conference, Cork, Ireland, Vol. 12, No. 1197, 27 Aug. – 2 Sept.
- Youssef, J., Matar, J., Rahme, P., Bou-Mosleh, C., 2016. A nearshore heaving-buoy sea wave energy converter for power production. In: International Conference on Sustainable Design, Engineering and Construction, Procedia Engineering, 145. Elsevier, pp. 136–143. doi:10.1016/j.proeng.2016.04.032.



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